Do Environmental Provisions in Trade Agreements Reshape Export Composition?

Evidence from the Universe of Chinese Customs Transactions, 2000–2015

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Abstract

Environmental Provisions (EPs) have been increasingly embedded in Preferential Trade Agreements (PTAs) over the last two decades, with the expectation that they will play an even more important role in future agreements. However, empirical evidence on their trade effects relies almost entirely on aggregate bilateral flows. This paper asks whether EPs can shift export composition towards green goods and away from dirty goods, using a firm-product-destination-year panel for the universe of Chinese customs transactions spanning from 2000 to 2015. A triple-difference design compares green and dirty products against neutral counterparts within each firm-destination-year combination, with firm-destination-year fixed effects accounting for everything constant within each firm-market-year triple. The results indicate that there is no effect on the green margin, while the dirty margin is less clear-cut. The inference on the green margin is robust to wild cluster bootstrap and permutation inference; that for the dirty margin is not. What emerges is that the content of environmental provisions matters more than their mere presence. The PTAs signed by China over 2000–2015 contained mostly cooperation language, which makes our results consistent with the existing literature, where significant effects are driven by specific and enforceable clauses rather than loose language.

1 Introduction

Over the last decades, the number of preferential trade agreements (PTAs) has increased substantially, and many now include environmental provisions (EPs) or other forms of environmental language (Mattoo, Rocha, and Ruta, 2020; Morin, Pauwelyn, and Hollway, 2017). Because climate change is likely to become an increasingly important political concern, and given the evidence that global trade may play a role in environmental decay (Zhang et al., 2017), environmental content in trade agreements is expected to feature more frequently and more prominently in trade policy, as it may serve as a possible tool to address the challenges posed by climate change (Frankel, 2009). Accordingly, the relationship between trade and the environment has attracted growing attention in the literature (Copeland, Shapiro, and Taylor, 2022), yet whether environmental provisions in trade agreements have a genuine effect on trade flows is still an open empirical question (Felbermayr, Peterson, and Wanner, 2025; Gutsch et al., 2025). This paper contributes to this literature by asking whether the environmental provisions included in preferential trade agreements signed by China between 2000 and 2015 shifted the composition of its exports toward greener products and away from dirtier ones, or whether those provisions left trade flows largely unaffected.

Most of the existing literature on the trade effects of environmental provisions in trade agreements comes from studies relying on aggregate bilateral trade flows, which makes it difficult to understand whether EPs cause a compositional shift in traded goods. In what the authors claim to be the first article to investigate the trade effects of environmental provisions in PTAs, Brandi et al. (2020) find that EPs reduce dirty exports and increase green exports when regulations are specific and enforceable, especially for developing countries. On the other hand, Berger et al. (2020) use a gravity panel to show that membership in PTAs with more environmental provisions is associated with lower bilateral exports overall, with the effect being concentrated in South-North trade flows and not robust to PPML. Specific and enforceable clauses are also those that drive the effect in Abman, Lundberg, and Ruta, 2024, who find that forest preservation and biodiversity clauses in PTAs reduce deforestation in partner countries relative to those that do not include such clauses. Taken together, the evidence suggests that it is the content of the specific provisions and chapters that matters, rather than their mere presence in trade agreements. This paper brings the question to the largest exporter in the world, asking whether the environmental provisions in China's PTAs between 2000 and 2015 had any effect on the composition of its exports, and whether the content of those provisions matters for trade flows.

Trying to answer this empirical question comes with some challenges. The first and most obvious is that countries do not sign trade agreements at random, and the signing of trade agreements is likely to be correlated with pre-existing relationships of some sort. Therefore, any comparison of export flows to PTA partners versus non-partners risks confounding the effect of environmental provisions with pre-existing differences in trade relationships. A second issue is that aggregate bilateral trade flows cannot distinguish a reallocation of export composition toward greener goods from an expansion or contraction of trade that happens to coincide with agreement entry into force. To address the empirical question, this study uses the universe of Chinese customs transactions between 2000 and 2015, a 45-million-observation firm-product-destination-year panel with granularity at the HS6 product level, which makes it possible to track the same firm exporting the same product to multiple destinations simultaneously, and to observe how those flows diverge when one destination is covered by a preferential agreement and another is not. This design isolates the effect of environmental provisions on trade composition from other confounding factors.

Over the study period, China signed 14 PTAs with 25 destination economies, with environmental content ranging from a handful of generic cooperation clauses to dedicated chapters. Data on the content of the environmental provisions come from two independent sources, the World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) and the TREND database (Morin, Pauwelyn, and Hollway, 2017). The former provides a count of environmental-law provisions in force with each destination at each point in time, while the latter provides a more detailed coding of the content of those provisions, including sub-indices covering areas such as green market access and enforcement stringency. Using two independent codings guards against idiosyncrasies of either scheme and allows to check for robustness of the results. We identify green goods on the basis of the OECD Combined List of Environmental Goods (CLEG) (Sauvage, 2014), while dirty goods are identified through a standard pollution-intensity classification that maps polluting industries to HS6 product codes, as developed in Mani and Wheeler, 1998. Everything else serves as the neutral comparison group.

These three product categories, green, dirty and neutral, represent the foundation of the empirical strategy. The empirical design applied in this study is a comparison of how the exports of green and dirty products react to environmental provisions, relative to neutral products, within the same firm, destination and year. A triple-difference with a rich set of fixed effects, including firm-destination-year fixed effects, absorbs everything that is constant within a firm's relationship with a specific destination market in a given year, including the agreement itself and any aggregate demand or supply shocks at the bilateral level. Because EP enter into force with the agreement itself and rarely change their depth or content afterwards, EP depth and PTA depth are highly correlated, as agreements with more EP content tend to be also deeper agreements overall. Therefore, all specifications control for PTA depth separately, so that the coefficient on environmental provisions reflects proper environmental content. In this design, identification comes from the differential sensitivity of green and dirty products to environmental provisions, relative to neutral products, which serve as a within-cell control since they are not expected to be affected by environmental provisions as much as green and dirty products are.

The results indicate that there is no effect on the green margin, so the export of green goods relative to neutral ones is not affected by the depth of environmental provisions in China's PTAs. This result is stable across six different estimation designs that vary the unit of observation, the measurement instrument, and the comparison group. The baseline uses the full firm-product-destination-year panel; a collapsed specification aggregates the same data to HS6-destination-year cells; and both are replicated using the TREND provision count as an alternative measure of EP depth to that offered by the World Bank Deep Trade Agreements database. Further robustness checks restrict the sample to CEM-matched destinations and to the common-support overlap region. Across all specifications, we find no statistically significant effect of environmental provisions on the export of green goods relative to neutral ones. Furthermore, these results are robust to wild cluster bootstrap inference (Cameron, Gelbach, and Miller, 2008) and permutation inference (Young, 2019). On the other hand, results for the dirty margin are less clear-cut. While we find no effect in the full panel, the collapsed panel shows a negative and significant coefficient, suggesting that while EPs may not change firm-specific behaviour, they may reduce the overall market-level export of dirty goods relative to neutral ones. However, this result does not survive more robust inference. Wild cluster bootstrap raises the p-value from below 0.001 to 0.07; a permutation test, which asks whether the same pattern would emerge by chance if EP depth were randomly reassigned across destinations, raises it further to 0.28.

The findings show that environmental provisions in China's PTAs between 2000 and 2015 had no measurable effect on export composition at the firm-product-destination-year level. This result, however, should not be taken as conclusive evidence that EPs in trade agreements have no effect on trade flows at all, but rather reflects the nature of the specific provisions themselves. During the period under analysis, the environmental content of China's agreements consisted almost entirely of soft cooperation language, rather than specific and enforceable clauses backed by dispute settlement mechanisms. The types of clauses that have been shown to produce significant trade outcomes in other contexts, such as tariff concessions on environmental goods and binding emissions or process standards, are largely absent from these agreements. In this sense, the null result is not a puzzle, but rather the expected outcome given the existing literature's own heterogeneity results, applied to a setting where the provisions lack the specificity and enforceability that drive the effects documented elsewhere. The fact that the null result is robust to a variety of estimation designs and inference methods further supports the conclusion that the lack of effect is not due to methodological shortcomings, but rather reflects the substantive nature of the environmental provisions in China's PTAs during this period. Moreover, the findings are informative for policy makers, as they suggest that the mere inclusion of environmental provisions in trade agreements may not be enough to achieve the desired trade outcomes if those provisions are just loose cooperation language, even in the case of the largest exporter in the world. Instead, the design and content of the provisions themselves, including specificity and enforceability, are likely to be crucial for achieving meaningful trade effects.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and the agreements. Section 4 presents the empirical strategy. Section 5 reports the results, with robustness analysis integrated as subsections. Section 6 concludes.

2 Related Literature

A large body of evidence documents that preferential trade agreements increase trade between member countries. Using a panel gravity model that addresses PTA endogeneity, Baier and Bergstrand (2007) estimate that an FTA roughly doubles bilateral trade over a ten-year period, a finding substantially larger than earlier cross-sectional estimates — as the authors themselves note — that did not control for selection into agreements. Subsequent work shows that the overall trade effect depends on agreement characteristics: the magnitude varies with depth and enforceability rather than being a uniform response to preferential market access (Baier and Bergstrand, 2009). A survey by Freund and Ornelas (2010) argues that the distinction between shallow agreements, which primarily reduce tariffs, and deep agreements, which extend to regulatory harmonization and other behind-the-border provisions, is central to understanding their economic effects.

This distinction has been formalized through provision-level coding efforts. The World Bank's Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) classifies over 280 agreements into dozens of policy areas, revealing that agreements with similar headline coverage can differ dramatically in substantive content. The DESTA depth index (Dür, Baccini, and Elsig, 2014) provides an independent coding across seven policy areas, confirming the same heterogeneity. Drawing on the World Bank Deep Trade Agreements database, Mattoo, Mulabdic, and Ruta (2022) find that deep agreements generate more trade creation and less trade diversion than shallow ones, and that some deep provisions have a public-good character that increases trade even with non-members. Taken together, these results point to a consistent conclusion: the effects of trade agreements depend critically on *what* is inside them. Whether the same holds for environmental clauses is the question this paper asks.

The interaction between environmental regulation and trade is organized around two mechanisms. The pollution haven hypothesis (Copeland and Taylor, 1994) predicts that stricter regulation erodes comparative advantage in dirty goods, shifting pollution-intensive production toward countries with weaker standards. The Porter hypothesis (Porter and Linde, 1995) predicts the opposite: well-designed regulation can trigger innovation and improve competitiveness. The empirical record on the first mechanism is mixed: while Antweiler, Copeland, and Taylor (2001) find that the income-driven technique effect of trade dominates the composition effect at the aggregate level, Levinson and Taylor (2008) instrument for abatement costs and show that industries facing the largest compliance cost increases see the largest rise in net imports — consistent with the pollution haven prediction once one addresses the econometric difficulties — chiefly endogenous abatement costs and omitted environmental regulation — that limited earlier studies. The dirty-industry classification used here draws on the pollution-intensive sector taxonomy introduced by Mani and Wheeler (1998) and Low and Yeats (1992).

Whether environmental provisions embedded in trade agreements can produce effects analogous to those of unilateral environmental regulation has been addressed by a small but growing literature. Two recent surveys reach a similar conclusion: the evidence on the effects of environmental provisions in trade agreements remains thin, and the results are highly dependent on the type of provision and the empirical setting (Felbermayr, Peterson, and Wanner, 2025; Gutsch et al., 2025). A few empirical studies point in the same direction.

Brandi et al. (2020) address a question closely related to the one posed in this paper. Combining the TREND environmental provision database (Morin, Pauwelyn, and Hollway, 2017) with bilateral trade flows from UN Comtrade for a panel of over 180 countries, they find that only provisions with a direct trade mechanism produce measurable effects: trade-restrictive provisions reduce dirty exports by about 5 percent of the mean; liberal provisions raise the green export share; general cooperation language shows no effect. Berger et al. (2020), using a similar gravity framework, find that PTAs with more environmental provisions are associated with lower bilateral exports, with the effect concentrated in South-North flows. The difference between these results comes down to provision heterogeneity: Brandi et al. (2020) decompose environmental content into distinct categories, while Berger et al. (2020) use an aggregate count that combines provisions with and without a trade mechanism into a single index, which masks the heterogeneity that drives Brandi et al.'s result.

Abman, Lundberg, and Ruta (2024) exploit within-agreement variation across environmental chapters: comparing agreements with and without specific forest preservation clauses, using satellite-based deforestation data, they find that these clauses offset the increase in deforestation that typically follows PTA entry into force. The finding demonstrates that environmental provisions can produce real environmental effects — but only when they contain a specific and identifiable mechanism. On the methodological side, Cherniwchan (2017) uses within-plant US manufacturing data to show that emissions changes under NAFTA operate through individual producers' behavior rather than market selection, demonstrating the value of micro-level data for this question.

A recent strand of work has brought the question specifically to China. Yue and Lin (2024) find that environmental provisions in China's PTAs inhibit exports of energy-consuming products. Zhu and C. Sun (2026) use the TREND database and report that EPs reduce dirty exports and increase green exports. Zhu and C. Sun (2025) document that environmental provisions are associated with quality improvements in Chinese firms' exports and increased imports of intermediate goods. The contrast with Zhu and C. Sun (2026), whose customs data overlap with those used here, deserves to be explicit. Their finding is identified from variation *across* firms and destinations — precisely the channel that firm-destination-year fixed effects absorb in the present design. The two sets of results are not in contradiction: Section 5.4 shows that moving from the across-firm comparison to the within-firm comparison shrinks the dirty coefficient by a factor of 2.7, a direct measure of how much of the aggregate association reflects between-firm composition rather than reallocation within the firm.

What emerges from this body of work is a consistent pattern: measurable effects on trade composition arise only when environmental provisions contain specific, enforceable clauses with an identifiable trade mechanism. General cooperation language, which characterizes the vast majority of chapters, does not produce detectable effects in studies that distinguish among provision types. This paper occupies a symmetric position relative to Brandi et al. (2020): the Chinese agreements in the period under study are dominated by exactly the kind of cooperation language that produced no effect in their decomposition. The present null is therefore the predicted outcome of the existing literature's own heterogeneity findings, applied to a sample where the treatment lacks the specificity that drives effects documented elsewhere.

3 Data

The analysis combines two data structures. On the treatment side, we draw on the environmental content of China's preferential trade agreements as coded independently by the World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) and by TREND (Morin, Pauwelyn, and Hollway, 2017). On the outcome side, we use the universe of Chinese customs transactions for 2000–2015 at the firm–HS6–destination–year level: 45.8 million observations after excluding Hong Kong and Macao, covering 462,651 exporting firms, 236 destination economies, and between 4,450 and 4,766 HS6 product codes depending on the year. The outcome variables are the logarithm of export value, the logarithm of export quantity, and the logarithm of unit value. Products are classified as green (OECD CLEG list), dirty (pollution-intensive sectors), or neutral. Table 1 defines the main variables used throughout the analysis.

Table 1: Variable descriptions

Variable	Notation	Description
Log export value	$\ln x^{val}_{fpdt}$	Logarithm of export value (USD)
Log export quantity	$\ln x^{qua}_{fpdt}$	Logarithm of export quantity
Log unit value	$\ln x^{uv}_{fpdt}$	Log value minus log quantity
Green product	g_p	= 1 if HS6 code is in the OECD CLEG list
Dirty product	b_p	= 1 if HS6 code belongs to a pollution-intensive sector
EP depth (WB)	EP^{WB}_{dt}	Count of environmental-provision categories (World Bank)
EP count (TREND)	EP^{TR}_{dt}	Count of individual environmental clauses (TREND)
Non-env. depth	TD_{dt}	Count of non-environmental provisions (World Bank)
Log tariff	τ_{pdt}	Logarithm of (1+ applied MFN tariff rate)
Log HHI (BACI)	HHI_{pdt}	Log Herfindahl index of destination-product concentration

Notes: Subscripts: f = firm, p = HS6 product, d = destination, t = year. Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

3.1 Chinese PTAs and their environmental content

Between 2000 and 2015, China signed preferential agreements covering 25 destination economies, of which 23 enter the main estimation sample after excluding Hong Kong and Macao (Table 2). Three features of this treatment landscape shape everything that follows.

First, the 23 treated destinations (listed in Appendix Table 25) correspond to only 13 distinct EP profiles—combinations of depth and entry timing. Ten destinations are parties to the ASEAN–China agreement; Laos also entered through the Bangkok Agreement and Singapore also signed a separate bilateral, leaving nine destinations that share the same ASEAN-only profile. Standard errors clustered by destination and the permutation inference described in Section 4.2 account for this structure.

Second, EP content is fixed at signature and almost never changes afterwards. Only three destinations (Laos, Singapore, and South Korea) exhibit within-country variation over time, and in each case depth rises, never falls. There is no treatment reversal in the sample.

Third, Hong Kong and Macao (CEPA, 2003) are entrepôt economies. They account for 24.4% of treated observations and 50.1% of treated export value, but a large fraction of what

Table 2: Chinese PTAs in force, 2000–2015

Agreement	Destinations	In force	WB EP	TREND EP
Bangkok Agr./APTA	Bangladesh, India, S. Korea, Laos, Sri Lanka	2002	1	1
ASEAN–China	Brunei, Cambodia, Indonesia, Malaysia, Myanmar, Philippines, Singapore, Thailand, Timor-Leste, Vietnam	2005	6	4
China–Chile	Chile	2006	5	30
China–Pakistan	Pakistan	2007	3	12
China–New Zealand	New Zealand	2008	7	53
China–Singapore	Singapore	2009	7	15
China–Peru	Peru	2010	12	47
China–Costa Rica	Costa Rica	2011	4	50
China–Iceland	Iceland	2014	6	28
China–Switzerland	Switzerland	2014	14	53
China–Australia	Australia	2015	3	24
China–S. Korea	South Korea	2015	17	72
Excluded from main specification				
CEPA Hong Kong	Hong Kong	2003	4	5
CEPA Macao	Macao	2003	5	5
Total treated destinations (excl. HK–MO)			23	

Notes: Entry-into-force years from the World Bank Deep Trade Agreements database and TREND. WB EP = count of environmental-provision categories (World Bank); TREND EP = count of individual environmental clauses (TREND). Values shown are the maximum observed over the sample period. Three destinations experience within-country increases in EP depth: Laos (Bangkok 2002 → ASEAN 2005), Singapore (ASEAN 2005 → bilateral 2009), and South Korea (Bangkok 2002 → bilateral 2015). Laos is listed under the Bangkok Agreement, its first agreement; it is also a party to ASEAN–China from 2005, so eleven destinations are ASEAN–China parties in total. Timor-Leste is assigned to the ASEAN–China agreement in the country lists used to expand agreements into destination–year rows, although it acceded to ASEAN only in 2022. It accounts for 0.02% of observations, and excluding it moves the dirty coefficient by less than 10^{-6} (leave-one-out exercise). Source: World Bank DTA Database; TREND Database.

the customs data record as Chinese exports to these destinations are goods transshipped and re-exported to third countries at a markup (Feenstra and Hanson, 2004). Including them would mix the demand of the true final destination with the bilateral fixed effects designed to absorb it, since the recorded destination and the actual final market differ systematically.[1]

Entry into the treated group is concentrated in two large cohorts (2002 and the 2005 ASEAN accession); most subsequent years add one destination at a time. Of the 236 destinations in the panel, 23 are ever treated and 213 are never treated within the sample window.

EP content is measured with two independent codings. The World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) yields EP_{dt}^{WB}, the count of environmental-provision categories in force with destination d at time t, and a non-environmental depth control TD_{dt} (total non-environmental provisions). TREND (TRade and ENvironment Database; Morin, Pauwelyn, and Hollway, 2017) yields EP_{dt}^{TR}, a count of individual environmental clauses, and sub-indices covering green market access, enforcement stringency, and hard versus soft provisions. Using two codings independently constructed from different frameworks guards against the idiosyncrasies of either; as shown below, they agree on the main findings.

Figure 1 traces the expansion of China's PTA network and the evolution of EP content over the sample period. The early agreements (Bangkok 2002, ASEAN 2005) contain minimal environmental language; substantive environmental chapters appear only from New Zealand (2008) onward, with the deepest content in the Switzerland and South Korea agreements (2014–2015).

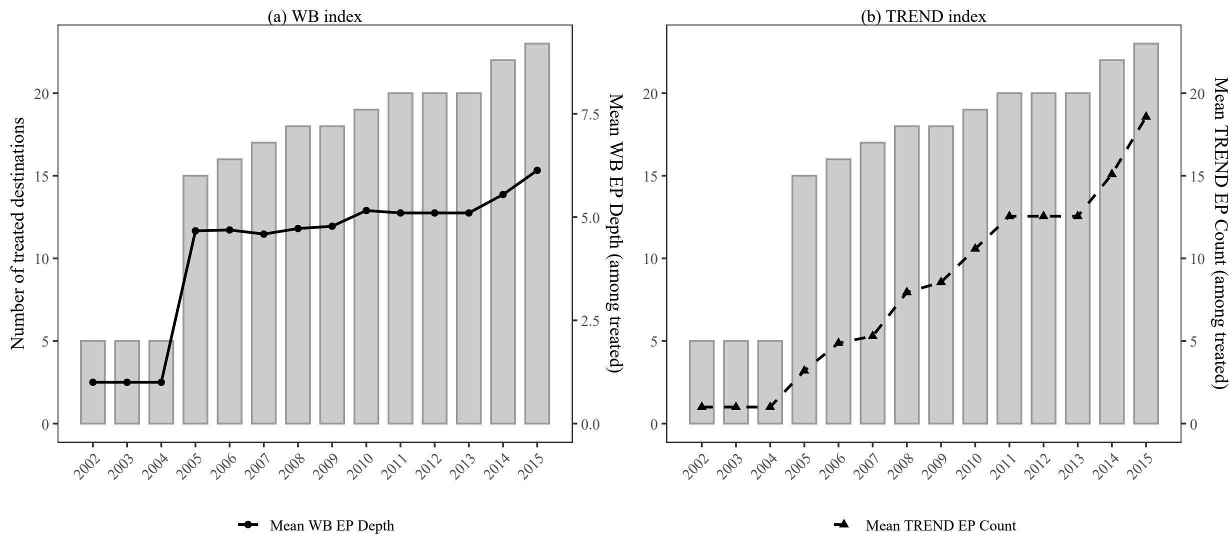
3.2 Customs data and product classification

The trade data are the universe of Chinese export customs transactions for 2000–2015, sourced from the Chinese Customs Office. Transactions are aggregated to the firm–HS6–destination–year level, yielding 49.2 million observations (45.8 million once Hong Kong and Macao are excluded), 462,651 exporting firms, 236 destinations, and between 4,450 and 4,766 HS6 product codes depending on the year. Three outcome variables are used throughout: the logarithm of export value ($\ln x_{fpdt}^{val}$), the logarithm of export quantity ($\ln x_{fpdt}^{qua}$), and the logarithm of unit value ($\ln x_{fpdt}^{uv}$, defined as value divided by quantity). Export value is the primary outcome; quantity and unit value serve as decomposition checks.

Green products are the 248 HS6 codes of the OECD Combined List of Environmental Goods (CLEG), as classified in Sauvage (2014). The customs panel uses the HS1996 vintage while the OECD list is defined in HS2012, so the list was harmonized. Of the 248 codes, 246

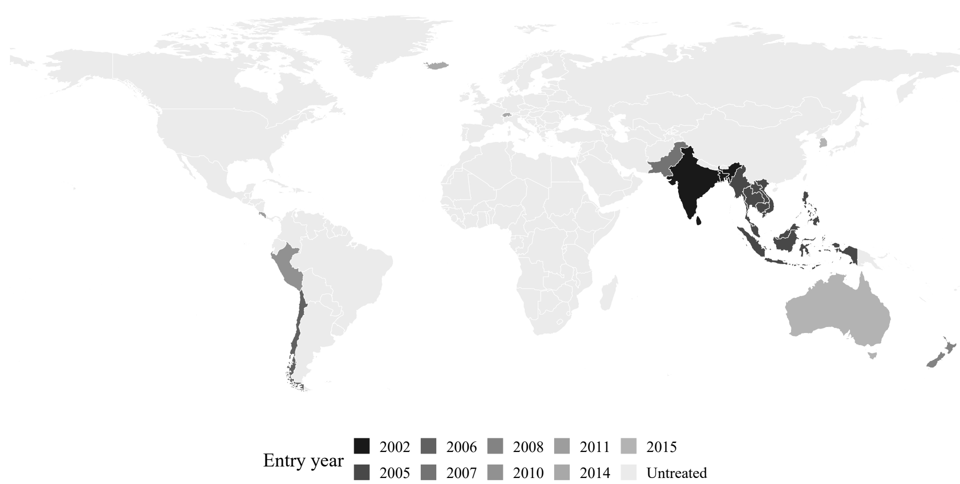
[1]Results including Hong Kong and Macao are reported in Section 5. Point estimates are substantively unchanged; the main effect is wider confidence intervals due to the additional noise from entrepôt flows.

Figure 1: Expansion of China's PTA network and EP content, 2000–2015



Notes: Bars show the number of treated destinations (excl. HK–MO) with positive EP depth in each year. Lines show the mean EP index value among treated destinations: WB EP Depth (solid circles, left panel) and TREND EP Count (dashed triangles, right panel). The two indices are plotted on separate axes because they differ in scale by a factor of approximately three by 2015. Source: World Bank DTA Database; TREND Database.

Figure 2: Geographic coverage of Chinese PTAs, 2000–2015



Notes: Shaded countries are destinations covered by a Chinese PTA with environmental provisions during the sample period. Shading indicates the year of entry into force. Hong Kong and Macao (excluded from the main specification) are not shown. Source: World Bank DTA Database.

have a one-to-one mapping to HS1996 and are translated directly. The remaining two (codes 871411 and 871419, which correspond to a single HS1996 heading split into two sub-codes in HS2012) are retained at the original HS1996 level; these two codes jointly account for a negligible share of green-goods trade, so no information of consequence is lost. [2]

Dirty products belong to the classic pollution-intensive sectors of Mani and Wheeler (1998) and Low and Yeats (1992): pulp and paper, industrial chemicals, petroleum refining, iron and steel, and non-ferrous metals, with cement added in an extended variant.[3]

The remaining HS6 codes form the neutral comparison group. Summary statistics appear in Table 3.

Table 3: Summary statistics (full panel, excl. HK–MO)

Variable	N	Mean	Median	Std. Dev.	Range
$\ln x^{val}_{fpdt}$	45,781,181	9.264	9.360	2.546	[0, 23.3]
$\ln x^{qua}_{fpdt}$	45,470,053	7.312	7.595	3.213	[−6.9, 25.7]
$\ln x^{uv}_{fpdt}$	45,470,040	1.986	1.609	2.306	[−13.0, 20.4]
g_p (green)	45,781,211	0.115	0	0.319	{0, 1}
b_p (dirty)	45,781,211	0.070	0	0.255	{0, 1}
EP^{WB}_{dt}	45,781,211	0.966	0	2.383	[0, 17]
EP^{TR}_{dt}	45,781,211	2.051	0	8.165	[0, 72]
τ_{pdt}	41,117,073	1.658	1.792	1.027	[0, 7.3]
HHI_{pdt}	45,306,336	−1.540	−1.610	0.592	[−3.3, 0]

Notes: 45,781,211 firm–HS6–destination–year observations after excluding Hong Kong and Macao. Observation counts differ across variables due to missing tariff data and quantity/unit-value availability. Mean, median, and standard deviation omitted for binary variables with more than two categories. Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Figure 3 plots the share of green and dirty products in total export value over time, separately for treated and untreated destinations. The two groups track each other closely throughout the sample period, with no visible divergence after agreements enter into force — consistent with the regression results reported below.

[2]As a check on the classification itself, the specification was re-estimated restricting green products to the 54 codes of the APEC Environmental Goods List (Sauvage, 2014), on which there is explicit multilateral political consensus and which form a complete subset of the OECD list. With the WB index the green coefficient flips sign to +0.0050 (s.e. 0.0127, $p = 0.69$); with TREND it moves from +0.0018 to +0.0032 ($p = 0.13$). Both remain indistinguishable from zero, with standard errors roughly double, as expected from restricting to 22% of the original green sample. Appendix Table 28 reports the full comparison.

[3]Products are mapped from ISIC Rev.2 to HS1996 using the official WITS/UNSD concordance, yielding 1,139 dirty HS6 codes. Seventeen codes that appear in both lists are assigned to the green category; the two lists are mutually exclusive.

Table 4: Summary statistics (collapsed panel, excl. HK–MO)

Variable	N	Mean	Median	Std. Dev.	Range
$\bar{y}_{pdt}$ (mean $\ln x^{val}$)	3,773,498	8.974	9.029	1.957	[0, 20.7]
Cell count (n_{pdt})	3,773,498	12.13	3	43.64	[1, 8,946]
EP_{dt}^{WB}	3,773,498	0.669	0	1.980	[0, 17]
EP_{dt}^{TR}	3,773,498	1.438	0	6.828	[0, 72]
g_p (green)	3,773,498	0.084	0	0.278	{0, 1}
b_p (dirty)	3,773,498	0.140	0	0.347	{0, 1}

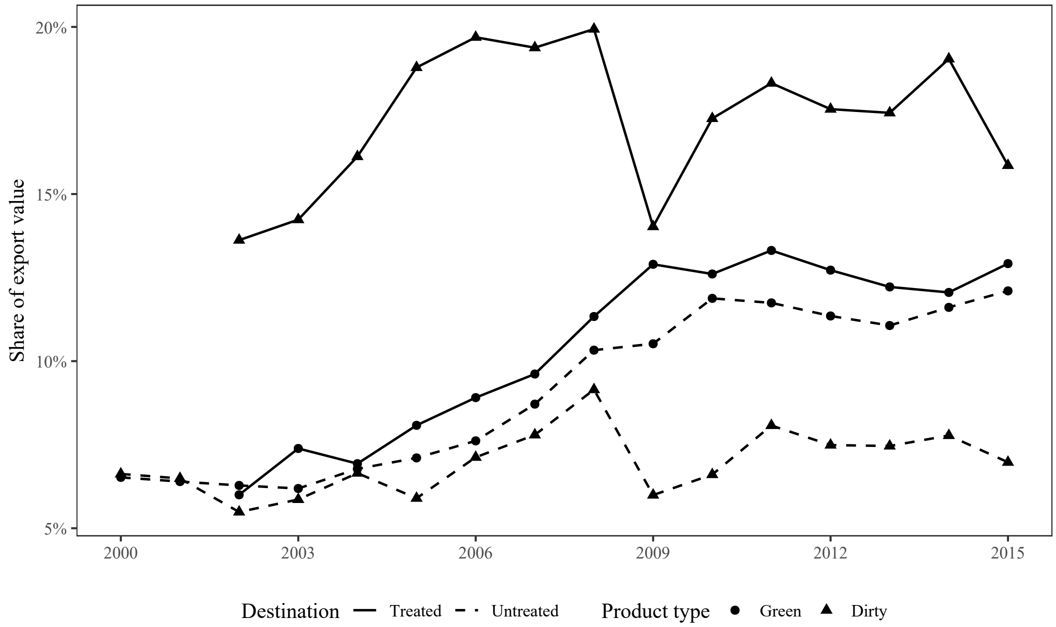
Notes: 3,773,498 HS6–destination–year cells. $\bar{y}_{pdt}$ is the within-cell mean of log export value across firms. Cell count n_{pdt} is the number of firm-level observations in each cell (used as regression weight). Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 5: Summary statistics (combined: full and collapsed panels)

	Full panel			Collapsed panel		
Variable	N	Mean	S.D.	N	Mean	S.D.
$\ln x^{val}$ / $\bar{y}_{pdt}$	45,781,181	9.264	2.546	3,773,498	8.974	1.957
$\ln x^{qua}$	45,470,053	7.312	3.213	—	—	—
$\ln x^{uv}$	45,470,040	1.986	2.306	—	—	—
g_p (green)	45,781,211	0.115	0.319	3,773,498	0.084	0.278
b_p (dirty)	45,781,211	0.070	0.255	3,773,498	0.140	0.347
EP_{dt}^{WB}	45,781,211	0.966	2.383	3,773,498	0.669	1.980
EP_{dt}^{TR}	45,781,211	2.051	8.165	3,773,498	1.438	6.828
τ_{pdt}	41,117,073	1.658	1.027	—	—	—
HHI_{pdt}	45,306,336	−1.540	0.592	—	—	—
Cell count (n_{pdt})	—	—	—	3,773,498	12.13	43.64

Notes: Full panel: 45,781,211 firm–HS6–destination–year observations. Collapsed panel: 3,773,498 HS6–destination–year cells. Both exclude Hong Kong and Macao. The green share is higher in the full panel (11.5% vs. 8.4%) because green products are exported by more firms per product–destination cell; conversely, the dirty share is higher in the collapsed panel (14.0% vs. 7.0%) because dirty products have fewer exporting firms per cell. Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Figure 3: Green and dirty export shares by treatment status, 2000–2015



Notes: Share of export value accounted for by green and dirty products, computed separately for destinations with positive EP depth (treated) and destinations with no PTA or no EP content (untreated) in each year. Excludes Hong Kong and Macao. Source: Chinese Customs Office.

3.3 Estimation samples

The analysis rests on four estimation objects built from the same underlying panel; their sizes and fixed-effect structures are summarized in Table 6.

The *full firm-level panel* is the main estimation object, with 45.8 million observations at the firm–HS6–destination–year level. The outcome variables are $\ln x^{val}_{fpdt}$, $\ln x^{qua}_{fpdt}$, and $\ln x^{uv}_{fpdt}$; the main fixed effects are firm–product–destination (θ_{fpd}), firm–destination–year (θ_{fdt}), and product–year (θ_{pt}). Standard errors are clustered by destination.

The *collapsed panel* aggregates the firm dimension, yielding 3.77 million HS6–destination–year cells. The outcome in each cell is the mean of log exports across the underlying firm-level observations, and regressions are weighted by the cell count (the number of firm-level observations in each cell). Fixed effects are product–destination (pd), destination–year (dt), and product–year (pt). The collapsed panel serves two purposes: it is a computational workhorse for the bootstrap and permutation inference battery, and it provides an internal cross-check against the full panel.[4] The two panels agree on the green margin; Section 5.4

[4]With 45.8 million observations and three sets of high-dimensional fixed effects, the full panel is computationally intensive, and the bootstrap requires several hours per specification. The collapsed panel reduces dimensionality by a factor of twelve, making the full inference battery feasible on a single machine. The equivalence between the two implementations is verified algebraically and numerically in Appendix A.

Table 6: Descriptive statistics

Full firm-level panel (excl. HK–MO)	
Observations (firm–HS6–dest–year)	45,781,211
Exporting firms	462,651
HS6 products	4,450–4,766
Destinations	236
of which treated (EP depth >0)	23
Distinct EP profiles (depth $\times$ timing)	13
Share of observations: green products	11.5%
Share of observations: dirty products	7.0%
Share of observations under an EP agreement	20.3%
Collapsed panel (HS6–dest–year)	
Cells	3,773,498
of which green / dirty	8.4% / 14.0%
Estimation sample (post-singleton)	3,681,023
PPML zero-filled grid (HS6–dest–year, excl. HK–MO)	
Cells	8,179,904

Notes: Hong Kong and Macao excluded throughout (they account for 24.4% of treated observations and 50.1% of treated export value). Product classification as described in the text. Source: Chinese Customs Office.

explains why they diverge on the dirty margin.

The *zero-filled PPML grid* (8.2 million cells) is built from the full panel by first aggregating to HS6–destination–year cells and then adding product–destination pairs with no recorded trade in a given year, conditional on the pair having at least one positive flow elsewhere in the sample period.[5]

Because the triple-difference compares green and dirty products to neutral ones, the credibility of the comparison group can be probed directly. Four control-group subsamples, each tightening the comparison in the direction of a specific threat, are summarized in Table 7 and described below. The prefix C- denotes the control-group variant.

C-prod-HS4 restricts the neutral comparison group to products belonging to the same HS4 family as a green code (106 HS4 families, covering 20.5% of observations). This guards against the concern that neutral products are too dissimilar from green ones to serve as a valid counterfactual. It should be read alongside C-overlap: multi-product firms may reallocate

[5]This conditional zero-fill — rather than a complete cross-join of all products, destinations, and years — keeps the grid computationally tractable while capturing the relevant extensive margin. A zero-fill at the firm–product–destination–year level would generate billions of rows, making estimation computationally infeasible; a PPML without zero-fill would miss the extensive margin entirely, since the estimator's advantage lies precisely in the information contained in the zeros.

Table 7: Control-group subsamples

Subsample	Threat addressed
C-prod-HS4 neutral products in the same HS4 family as a green code	Neutral products differ from green/dirty
C-overlap products exported both to treated and untreated destinations	Extrapolation outside common support
CEM destinations destinations matched on year-2000 GDP p.c., growth, MFN tariff	Selection of partner countries
Deep vs. shallow PTA partners only, split at median EP depth	The "any-agreement" contrast itself

Notes: All observation counts refer to the full firm-level panel (excl. HK–MO). Source: Chinese Customs Office; World Ba Database; TREND Database.

toward core products when trade costs fall (Eckel and Neary, 2010), potentially displacing neutral products in the same HS4 family and biasing the contrast.

C-overlap retains only products exported to both treated and untreated destinations (98.5% of all HS6 codes, 21.5 million observations). This ensures that the identifying variation does not rely on products traded exclusively with one group of destinations, guarding against the possibility that multi-product firms reallocate across products within the same HS4 family when trade costs fall ().[6]

CEM destinations applies coarsened exact matching (Iacus, King, and Porro, 2012) at the destination level, matching treated destinations to untreated ones on year-2000 GDP per capita, GDP growth, and MFN tariff levels. The resulting sample contains 19 treated and 40 control destinations (approximately 14.0 million observations before singleton removal; 13,728,510 after iterative removal of singletons) and addresses the concern that treated and untreated destinations differ systematically on dimensions correlated with export composition.

Deep vs. shallow drops all untreated destinations and splits the remaining PTA partners at the median EP depth (16 deep, 7 shallow partner countries in the baseline sample; 5.3 million observations). This design asks whether composition differs between deep and shallow environmental chapters, conditional on having an agreement at all, and addresses the concern that any observed effect reflects the agreement rather than its environmental content.

A genuine composition effect should survive every change to the comparison group; an artifact will move when the comparison changes.

[6] In practice the restriction is close to non-binding: it removes 314 of the 21,519,511 observations in the estimation sample. The result confirms that no identifying variation comes from products traded with only one group of destinations.

4 Empirical Strategy

Three challenges shape the identification strategy: countries select into agreements, environmental and overall agreement depth correlate strongly, and the number of independent treatment units is small.

A starting point would be to estimate the level effect of environmental provisions on trade—regressing log exports on environmental depth with fixed effects. A structural argument shows why this is uninformative in this setting.

The argument is independent of the empirical results. Any fixed-effects structure that seriously addresses bilateral confounders requires a destination–year intercept, which absorbs *all* variation at the (d, t) level—including environmental depth itself. With only 13 distinct EP profiles and near-zero within-destination change over time, "environmental depth" and "having a deep agreement" are empirically indistinguishable. The interaction $EP_{dt} \times g_p$ survives the destination–year intercept because the product dimension varies *within* the (d, t) cell. This is why the specification is built on composition—the differential response of green and dirty products relative to neutral ones—rather than on levels.

This design cannot show that EP-heavy agreements affected aggregate trade levels — that effect is confounded with the agreement itself. The evidence speaks only to the composition margin.

4.1 The triple-difference specification

The question is whether environmental depth shifts the gap between green (or dirty) and neutral log exports among treated destinations.[7] OLS gives equal weight to each observation, so the coefficient is a per-observation average, not a per-firm or per-destination average.[8] The main specification is

$$\ln x_{fpdt} = \beta_1\, EP_{dt}{\times}g_p + \beta_2\, EP_{dt}{\times}b_p + \gamma_1\, TD_{dt}{\times}g_p + \gamma_2\, TD_{dt}{\times}b_p + \theta_{fpd} + \theta_{fdt} + \theta_{pt} + \varepsilon_{fpdt}, \quad (1)$$

where f indexes firms, p HS6 products, d destinations, and t years; g_p and b_p are the green and dirty product indicators defined in Section 3; EP_{dt} is environmental depth and TD_{dt} non-environmental agreement depth. Environmental depth enters the regression only through its

[7] In potential-outcomes terms, the target can be written as $\beta_1 = E\big[(y_{fpgdt}(1) - y_{fpgdt}(0)) - (y_{fpndt}(1) - y_{fpndt}(0)) \mid d, t \text{ treated}\big]$, where $y(1)$ and $y(0)$ denote log exports of green product g versus neutral product n under deeper and unchanged environmental content respectively, holding the agreement fixed.

[8] A firm exporting 100 products receives 100 times the weight of a firm exporting one product. The coefficient therefore reflects the composition of the estimation sample rather than a simple average across firms or destinations.

interactions with g_p and b_p, not on its own: the level effect of EP_{dt} is absorbed by θ_{fdt}, and what remains is the differential response of green and dirty products relative to neutral ones.

The three sets of fixed effects absorb different sources of confounding variation, and the choice of which to include is dictated by the structure of the treatment.

The firm–product–destination fixed effects θ_{fpd} absorb time-invariant features of each firm–product–market relationship—for example, a firm's long-standing contract with a South Korean buyer of solar panels, or an established network for exporting steel to a particular ASEAN destination.

The firm–destination–year fixed effects θ_{fdt} are what drives identification. They absorb all time-varying bilateral factors — entry into new markets, aggregate demand shifts, exchange rate movements — that affect a firm's exports to a destination uniformly across products. The agreement dummy, overall agreement depth, destination demand shocks, and the non-random selection of destinations into agreements all drop out. Identification comes from comparing green and dirty products to neutral ones *within* the same cell as environmental content varies across destinations and over time.

The product–year fixed effects θ_{pt} absorb global product-level shocks common to all destinations—for instance, the worldwide expansion of the solar panel market in the late 2000s, or global fluctuations in steel prices—so that these shocks do not contaminate the composition comparison.[9]

The three-way fixed-effect structure absorbs a large number of singletons — observations whose identifying variation is fully accounted for by at least one fixed effect. Iterative singleton removal (Correia, 2017) reduces the estimation sample from 45.8 million raw observations to 21,519,511 in the full panel (a 53% reduction) and from 3,773,498 cells to 3,681,023 in the collapsed panel.

Selection into agreements is not eliminated but substantially narrowed: θ_{fdt} absorbs the level effect, so selection contaminates β_1 only through differential pre-trends in green versus neutral exports across destinations. Section 5 tests this directly with destination-specific trends and finds no change in the coefficient.

The omitted category is neutral products. β_1 measures how green products move *relative to neutral ones* as environmental depth increases; β_2 measures the same contrast for dirty products. Each coefficient is already the difference from the neutral baseline. The green–dirty differential ($\beta_1 - \beta_2$) is a derived quantity, not the parameter the regression identifies, and should not be read as such.

The same World Bank non-environmental depth control is used in both the WB and the

[9]Firm–product–year fixed effects would be computationally infeasible given the panel size. Product–year effects are sufficient because firm–product shocks that do not vary across destinations are already absorbed by the combination of θ_{fpd} and θ_{fdt}.

TREND specifications: TREND does not code non-environmental provisions, so no TREND-based analogue exists.

The non-environmental depth interactions ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) separate environmental from general agreement content. TD_{dt} counts all non-environmental provisions, but not all of these are directly relevant to green or dirty trade; it is therefore an imperfect proxy for the non-environmental component of agreement depth that might independently affect export composition. An imperfectly measured control under-controls: β_1 absorbs whatever share of general-depth variation the proxy fails to capture.[10] Rather than assert a direction, Section 5 reports the green coefficient under four alternative depth controls and shows that the point estimate moves within a band narrower than one standard error (Table 14).

Collinearity between EP depth and non-environmental depth is substantial: the two correlate at $\rho = 0.91$ across treated destination–years, and the VIF for EP depth in the main specification is 5.8.[11] The TREND count is less collinear with non-environmental depth ($\rho = 0.50$), which is one reason the WB confidence intervals are wider. That two codings with such different degrees of overlap return the same null is reassuring, since a result driven by the inability to separate environmental from general depth would be expected to differ between them.

Standard errors are clustered by destination. EP depth varies across destinations and years but changes in only three destinations during the sample period, making treatment effectively destination-level. Clustering by destination treats all observations from the same country as a single cluster, which is the appropriate level when the treatment is in practice a destination-level assignment (Abadie et al., 2023).

A further identification concern works in a conservative direction. If deeper environmental chapters tend to be signed where green demand is already rising, EP_{dt} is positively correlated with an omitted destination-specific trend, which would bias β_1 positively. With a null coefficient, any upward bias from this channel implies the true effect is zero or negative—the null is a conservative rather than a lenient reading of the data.

Because environmental depth is a continuous dose switching on at staggered dates, the coefficient is a weighted average of dose- and cohort-specific composition responses, not a simple average treatment effect. Since the coefficient is near zero across specifications, the

[10] If the non-environmental depth control imperfectly measures the true confounding depth, $\hat{\beta}_1$ absorbs the part of general-depth variation the proxy misses. Because EP and total depth correlate at $\rho = 0.91$, this omitted component is positively associated with EP, so the bias pushes $\hat{\beta}_1$ upward. With a null green coefficient, any upward bias implies the true effect is zero or negative — the null is a conservative reading of the data.

[11] The Variance Inflation Factor measures how much collinearity inflates the variance of the estimated coefficient relative to an orthogonal design. A VIF of 5.8 is somewhat above the maximum of 4.6 that Brandi et al. (2020) report for their provision-type variables on a panel of 680 agreements.

choice of weighting scheme does not affect the conclusion: a weighted average of near-zero effects stays near zero regardless of the weights.[12]

4.2 Inference with few treated clusters

Of the destination clusters in the estimation sample, only 23 are treated, corresponding to 13 distinct EP profiles. With so few treated clusters, asymptotic cluster-robust standard errors are unreliable: the finite-sample approximation to the t-distribution is too optimistic, and p-values systematically over-reject the null (Cameron, Gelbach, and Miller, 2008; MacKinnon and Webb, 2017). Every headline estimate is therefore subjected to two additional tests.

Wild cluster bootstrap. The wild cluster bootstrap (Cameron, Gelbach, and Miller, 2008; Roodman et al., 2019) provides inference that does not rely on asymptotic critical values and is appropriate for settings with few clusters. In both panels, fixed effects are partialled out via the Frisch–Waugh theorem before the bootstrap is applied, because the bootstrap implementation requires a single set of absorbed effects. Point estimates are identical to those from the direct estimation; only computational cost differs.[13]

Permutation inference. The permutation test reshuffles EP profiles across the 23 treated destinations, testing whether the specific EP content — not the agreement itself — matters for composition.[14] With 1,000 replications, the test asks whether the observed coefficient is unusual relative to the placebo distribution.

The null hypothesis is that, given the set of 23 treated destinations, it does not matter which destination received which EP profile: the test permutes profiles among the treated, without adding or removing destinations from the treated group. Because depth and timing are permuted jointly, the test cannot separate the effect of environmental content from the

[12] Under continuous treatment entering at staggered dates, the TWFE estimator produces a weighted average of dose- and cohort-specific effects whose weights need not be convex (Callaway, Goodman-Bacon, and Sant'Anna, 2024; Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021). Two features limit what this costs in this setting. First, a weighted average of near-zero effects stays near zero unless the underlying effects are large and of opposite signs, of which the sub-index and leave-one-out evidence gives no indication. Second, the permutation test is agnostic about weighting: it asks whether the observed coefficient—whatever weighted average it represents—is unusual relative to placebos.

[13] Partialling out introduces two minor approximations. First, the bootstrap treats all fixed effects as if nested within the destination cluster. In the full panel this is nearly true: firm–product–destination and firm–destination–year are each nested within a destination; only product–year is not. In the collapsed panel the nesting is less complete (only destination–year is strictly nested), so the approximation is somewhat less tight. Second, the small-sample adjustment sees only the residualized regressors rather than the regressors plus the absorbed fixed effects. Neither approximation has practical relevance here: the conclusions are insensitive to the choice of panel, and the key dirty coefficient has a bootstrap p-value of 0.07 regardless of which approximation applies.

[14] The test operates only among treated destinations because permuting between treated and untreated destinations would test a different hypothesis (the effect of the agreement as such), which is unidentifiable as shown in Section 4.

effect of entry timing.[15]

Table 8 reports wild cluster bootstrap p-values and permutation p-values for the headline estimates.

[15]Estimates on the collapsed and destination-level panels were produced twice, in STATA (`reghdfe`, Correia, 2017; `boottest`, Roodman et al., 2019; `ppmlhdfe`, Correia, Guimarães, and Zylkin, 2020; `eventstudyinteract`, L. Sun and Abraham, 2021) and in R (`fixest`, `fwildclusterboot`); across 44 result files the two implementations agree on every point estimate to at least eight significant digits. Estimates on the 45.8-million-row firm-level panel exist only in STATA: `fixest` cannot complete them on the available hardware. They are validated instead by an algebraic identity—the firm-level panel estimated with $pd+dt+pt$ fixed effects reproduces the weighted collapsed-panel estimate to nine significant digits (Appendix A)—which checks the data construction of the two pipelines against each other. All reported numbers are the STATA ones. Two quantities legitimately differ between implementations and are discussed where they arise: bootstrap and permutation p-values, which carry Monte Carlo error, and the Sun–Abraham standard errors of Appendix C, which reflect a genuine methodological difference between `eventstudyinteract` and `fixest::sunab` (see Appendix C).

Table 8: Wild cluster bootstrap p-values (9,999 replications)

	(1)	(2)	(3)	(4)
Panel A — collapsed panel (WB index)				
EP depth × Green: coefficient	-0.00457	0.00173	-0.00433	-0.00675
Bootstrap p-value	0.649	0.825	0.546	0.484
EP depth × Dirty: coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Bootstrap p-value	0.072	0.005	0.047	0.195
Panel B — collapsed panel (TREND index)				
EP count × Green: coefficient	0.00181	0.00178	0.00161	0.00017
Bootstrap p-value	0.390	0.414	0.508	0.949
EP count × Dirty: coefficient	0.00035	-0.00001	-0.00025	0.00084
Bootstrap p-value	0.857	0.995	0.905	0.635
Panel C — full firm-level panel (WB index)				
EP depth × Green: coefficient	-0.00226	-0.00090	-0.00185	-0.00196
Bootstrap p-value	0.686	0.833	0.673	0.611
EP depth × Dirty: coefficient	-0.00435	-0.00409	-0.00559	-0.00574
Bootstrap p-value	0.185	0.176	0.049	0.035
Panel D — full firm-level panel (TREND index)				
EP count × Green: coefficient	-0.00010	-0.00011	-0.00012	-0.00039
Bootstrap p-value	0.931	0.919	0.920	0.703
EP count × Dirty: coefficient	-0.00089	-0.00101	-0.00140	-0.00140
Bootstrap p-value	0.177	0.142	0.069	0.082

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

Asymptotic p-values from destination-clustered standard errors are unreliable with 23 treated clusters (roughly 14 distinct EP profiles). The wild cluster bootstrap reconstructs the coefficient distribution by resampling 9,999 times and produces substantially more conservative p-values. The baseline dirty coefficient (Panel A, column 1) has asymptotic $p < 0.001$ but bootstrap $p = 0.072$.

Panels C and D computed via `boottest` in Stata on residualized data (Frisch–Waugh), because the command does not support more than one set of absorbed fixed effects.

5 Results

5.1 The green margin: a bounded null

Table 9 reports the main estimates. In the full firm-level panel, the EP×green coefficient is -0.0023 (s.e. 0.0039) with the WB index and -0.0001 (s.e. 0.0010) with TREND. The joint F-test on all four composition interactions is $p = 0.31$ (WB) and $p = 0.71$ (TREND). In the collapsed panel the WB estimate is -0.0046; wild cluster bootstrap $p = 0.65$; permutation $p = 0.60$. The observed green coefficient is unremarkable in the placebo distribution: $p = 0.60$ for the permutation test, meaning 60% of placebo permutations produce a coefficient at least as large in absolute value.

To calibrate magnitudes, consider the closest benchmark. Brandi et al. (2020) find that one liberal environmental provision raises the green share of developing-country exports by around 0.4 percentage points — roughly +0.16 log points in the metric of equation (1), assuming a baseline green share of about 2.5% — per provision. Section 4.2 argued that asymptotic inference is unreliable with 23 treated clusters; that argument extends to interval estimates, so the bounds cited here are wild cluster bootstrap intervals. The full-panel 95% bootstrap confidence interval for EP×green is $[-0.0353, +0.0355]$ per WB provision (collapsed: $[-0.0182, +0.0317]$). The corresponding asymptotic interval, $[-0.0100, +0.0055]$, is roughly six times narrower — reporting that level of precision would claim the kind of certainty this paper's own inference section declares unwarranted.

To interpret the bootstrap bounds: the upper end of the 95% interval for EP×green is +0.0355 per provision in the full panel. The Brandi et al. (2020) equivalent effect translates to roughly +0.16 log points per provision. The upper bound here is about a quarter of that figure, so the design can exclude effects at or above the Brandi level but cannot exclude smaller positive effects. A design powerful enough to bound the null more tightly would require more treated clusters. In per-provision terms, the interval $[-0.0353, +0.0355]$ spans about ±3.5%; in per-standard-deviation terms (one SD of WB EP depth in the estimation sample is approximately 2.4 provisions), the bounds imply at most a ±8–9% change in green relative to neutral exports. Appendix Table 29 reports the implied minimum detectable effects in both terms. Table 10 places the bounds alongside the Brandi et al. (2020) benchmark.

5.2 Stability across control groups

Across nine designs — collapsed panel with WB, collapsed panel with TREND, full panel with WB, full panel with controls, full panel excluding ASEAN, full panel including HK/Macao, C-prod-HS4, C-overlap, and CEM — the EP×green coefficient stays between -0.0002 and

Table 9: Triple-difference estimates: EP content and export composition

	WB EP depth		TREND EP count	
	(1)	(2)	(3)	(4)
	×green	×dirty	×green	×dirty
Panel A. Full firm-level panel				
EP interaction	−0.0023	−0.0044	−0.0001	−0.0009
	(0.0039)	(0.0022)	(0.0010)	(0.0006)
Asymptotic p	0.57	0.052	0.92	0.15
Bootstrap p	0.69	0.18	0.93	0.18
Firm–product–dest FE	Yes	Yes	Yes	Yes
Firm–dest–year FE	Yes	Yes	Yes	Yes
Product–year FE	Yes	Yes	Yes	Yes
Observations	21,519,511		21,517,666	
Clusters	225 destinations			
Joint F (p-value)	0.31		0.71	
Panel B. Collapsed panel (HS6–dest–year cells)				
EP interaction	−0.0046	-0.0119^{***}	+0.0018	+0.0004
	(0.0070)	(0.0030)	(0.0018)	(0.0016)
Asymptotic p	0.51	<0.001	0.32	0.83
Bootstrap p	0.65	0.07	0.39	0.86
Permutation p	0.60	0.28	0.16	0.82
Product–dest FE	Yes	Yes	Yes	Yes
Dest–year FE	Yes	Yes	Yes	Yes
Product–year FE	Yes	Yes	Yes	Yes
Cells	3,681,023			
Clusters	228 destinations			

Notes: All specifications include TotalDepth×green and TotalDepth×dirty controls (not shown). Wild cluster bootstrap: B = 9,999 (Cameron, Gelbach, and Miller, 2008; Roodman et al., 2019). Permutation: 1,000 reassignments of entire EP profiles across treated destinations. Bootstrap 95% CIs — Panel A (1)–(4): $[-0.035, +0.036]$, $[-0.043, +0.011]$, $[-0.0037, +0.0025]$, $[-0.0027, +0.0007]$; Panel B: $[-0.018, +0.032]$, $[-0.018, +0.002]$, $[-0.0017, +0.0071]$, $[-0.0031, +0.0056]$. $^{***}p < 0.001$ (asymptotic only; does not survive bootstrap or permutation — see Section 5.4).

Table 10: Benchmark against Brandi et al. (2020): equivalent effect and comparison with the confidence intervals

Margin	Brandi effect (log points)	Asy. CI upper	WCB CI upper/coef.	Ratio
Green (liberal)	0.1570	0.0055	0.0355	1/4 (WCB)
Dirty (restrictive)	-0.0513	-0.0044 (point est.)	–	1/12

−0.0046 and is never significant (Table 11).[16] The stability is the result: an artifact of the comparison group would move when the comparison group changes.

Table 11: Stability of the EP×green coefficient across estimation designs

Sample / design	Obs.	EP×green	Asymptotic p
Panel A — WB index			
Full panel (firm level)	21.5M	−0.0023	0.57
Collapsed panel	3.7M cells	−0.0046	0.51
Control-group subsamples			
C-prod-HS4 (within-HS4 neutral products)	3.8M	−0.0009	0.86
CEM-matched destinations	13.7M	−0.0023	0.54
Deep vs. shallow (PTA partners only)	5.3M	−0.0022	0.53
Common support (C-overlap)	21.5M	−0.0022	0.57
Full-panel variations			
With controls	19.3M	−0.0002	0.94
Excluding ASEAN	19.2M	−0.0025	0.48
Including HK–Macao	23.6M	−0.0009	0.80
Panel B — TREND index			
Full panel (firm level)	21.5M	−0.0001	0.91
Collapsed panel	3.7M cells	+0.0018	0.32
Control-group subsamples			
C-prod-HS4 (within-HS4 neutral products)	3.8M	+0.0005	0.68
CEM-matched destinations	13.7M	−0.0006	0.59
Deep vs. shallow (PTA partners only)	5.3M	−0.0004	0.72
Common support (C-overlap)	21.5M	−0.0001	0.91
Fixed effects as in baseline		Yes (all rows)	

Notes: Each row estimates equation (1) on the indicated subsample or design; asymptotic destination-clustered p-values. Fixed effects: $fpd+fdt+pt$ (full panel); $pd+dt+pt$ (collapsed). CEM matches on log GDP p.c., growth, and MFN tariff (year 2000). Deep/shallow split at the median of maximum EP depth: 16 deep vs. 7 shallow partner destinations (excl. HK/Macao; 9 shallow in the variant that includes them). Full-panel variations (With controls, Excluding ASEAN, Including HK–Macao) are not estimated for the TREND index. Controls: MFN tariff, HHI, antidumping indicator.

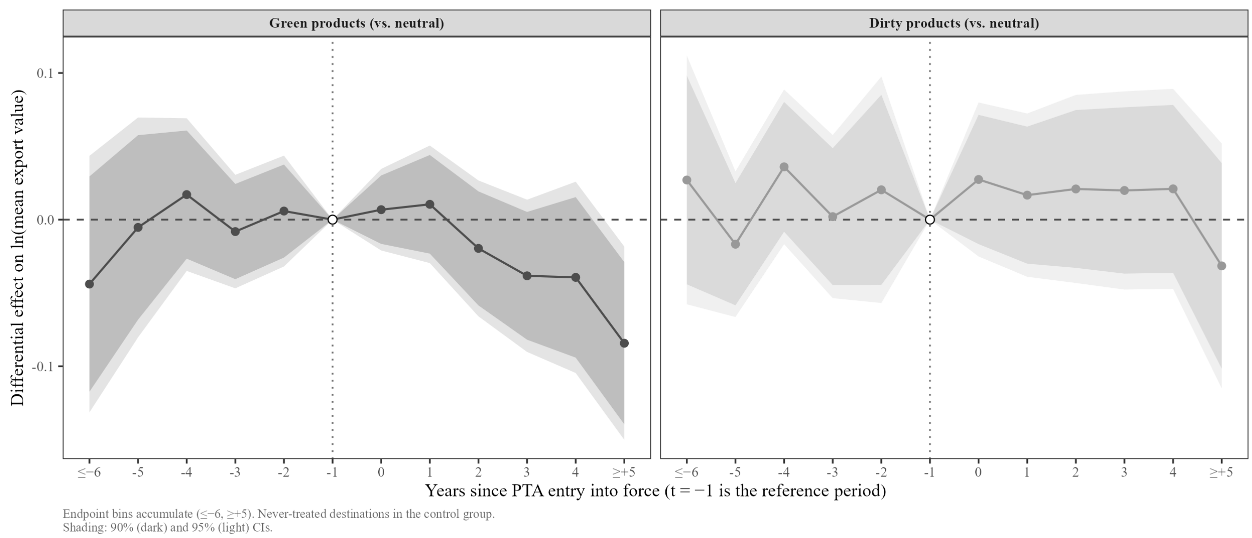
5.3 Dynamics

Figure 4 plots the event study of the green and dirty versus neutral gap around entry into force, using never-treated destinations as controls. Differential pre-trends are flat for both

[16] The full robustness battery in this paper runs approximately forty tests. Under independent testing, about two would appear significant at $p < 0.05$ by chance. The RegulatorySpace sub-index — the one instance that approaches significance in Table 23 — should be interpreted in this light.

margins. Before taking that as reassurance, a clarification is in order: parallel trends is a claim about the unobserved counterfactual path of treated destinations *after* treatment. A flat pre-trend rules out the most easily detectable violations — a composition gap already in motion before the agreement — but does not establish that the assumption holds exactly thereafter. The exercise is a falsification, not a proof.

Figure 4: Event study: green and dirty vs. neutral products around PTA entry into force



Notes: Collapsed panel; FE $pd + dt + pt$; destination-clustered 95% confidence bands; $t = -1$ is the reference period; endpoint bins accumulate. Source: Chinese Customs Office; World Bank DTA Database.

A mild negative drift for green products in the late post-period (bins $\geq +5$) is identified only by early cohorts (ASEAN 2005, Chile 2006, Pakistan 2007, New Zealand 2008). It disappears under the Sun–Abraham estimator applied to the destination-level composition gap: the aggregated ATT is -0.042 ($p = 0.27$) for the green gap and $+0.073$ ($p = 0.28$) for the dirty gap — consistent with early-cohort heterogeneity rather than any effect of treatment (L. Sun and Abraham, 2021). The Sun–Abraham estimator answers a narrower question than equation (1) — it binarizes treatment (EP present or absent, discarding depth), carries no depth control, and operates on a destination–year composition gap collapsed across products — so it should not be read as a replication. It is a diagnostic for staggered-timing and cohort heterogeneity, and that is how it is used here.

The full interaction-weighted event study is reported in Appendix C. With standard errors that account for the estimation uncertainty in cohort-share weights, no coefficient in the $[-10, +8]$ window is individually distinguishable from zero on either margin. Two or three nominally significant coefficients appear among 28 estimates, which is what sampling noise produces under exactly flat pre-trends with 23 treated clusters. The ATT is null across all three specification variants: $p = 0.28$ in the baseline, $p = 0.11$ in the $[-6, +5]$ window,

$p = 0.30$ excluding the 2014–2015 cohorts.

5.4 The dirty margin: anatomy of a false positive

The WB×dirty coefficient in the collapsed panel (-0.0119, asymptotic $p < 0.001$) deserves a subsection of its own. It illustrates, step by step, how a coefficient that looks robust under standard inference does not survive robust inference.

The two panels tell different stories on the dirty margin and the same story on the green margin. On green, both are indistinguishable from zero (-0.0046 collapsed, -0.0023 full panel). On dirty they differ by a factor of roughly 2.7 (-0.0119 collapsed versus -0.0044 full panel). The gap is a substantive difference, not a discrepancy: the collapsed panel compares products across all firms exporting to a given destination–year, so any shift in *which* firms export dirty goods enters the coefficient. The firm–destination–year fixed effects in the full panel remove this between-firm channel and isolate reallocation *within* the firm. Approximately three-fifths of the collapsed dirty coefficient reflects between-firm composition rather than within-firm response: $(0.0119-0.0044)/0.0119 = 0.63$. The full panel is therefore the reference specification for the dirty margin.

The wild cluster bootstrap raises the p-value to 0.07. Permutation inference on the exact estimating equation goes further: reshuffling entire EP profiles across the 23 treated destinations, 27.8% of placebo draws produce a coefficient as large in absolute value, giving $p = 0.28$ (Table 12). A coarser permutation on a destination–year–product-type aggregate reverses the sign entirely ($+0.005$, $p = 0.49$). The aggregation removes the within-cell heterogeneity across firms and product varieties that the disaggregated coefficient exploits, so the two tests are not answering exactly the same question, but they reach the same verdict. Sign instability under aggregation is itself evidence of fragility.

A leave-one-out exercise across the 23 treated destinations reveals the mechanism, and it is not the one the phrase "driven by a single country" usually implies. The *point estimate* is remarkably stable: it sits between -0.0097 and -0.0133 in every one of the 23 leave-one-out samples (Table 13). Dropping large treated destinations such as India or Pakistan moves it by 1 to 4% while leaving the standard error essentially unchanged. Those are not the pivotal observations.

What is not stable is the *precision*. Dropping Australia moves the coefficient by 13% (to -0.0103) but nearly triples its standard error, from 0.0030 to 0.0087: it is this, not the movement of the estimate, that carries the p-value to 0.24. Dropping South Korea doubles the standard error, giving -0.0097 with $p = 0.09$. Australia and South Korea are therefore not outliers exerting leverage on the estimate. These destinations provide the identifying variation — removing them leaves too little information for a precise estimate, rather than revealing a different effect. The coefficient is identified by a thin slice of variation concentrated in two destinations.

One might read the stability of the point estimate as evidence of a genuine distributed effect: if every exclusion produces a coefficient of similar magnitude, the effect does not rest on a single observation. This reading requires the estimate to survive robust inference, which it does not — the bootstrap consistently assigns $p = 0.07$–0.28 across specifications.

As an alternative depth control, this exercise also uses the DESTA index (Dür, Baccini, and Elsig, 2014), introduced in Section 2. The choice of depth control changes which destination's removal destabilizes the coefficient: with TotalDepth it is Australia, with DESTA it is South Korea. That the pivotal destination depends on a modelling choice unrelated to environmental content reinforces the fragility diagnosis. A genuine identified effect does not behave this way.

Australia and South Korea also happen to be among the destinations with the deepest EP coverage in the sample (WB depth 12 and 17, respectively). A dose-response reading — that the dirty effect is real but concentrated in the highest-dose observations — is not ruled out by the leave-one-out geometry alone. The DESTA depth control, which shifts the pivotal observation from Australia to South Korea, does not separate these two interpretations. We flag this ambiguity without resolving it.

The TREND index never shows the effect (bootstrap $p = 0.86$). In the full panel the coefficient is -0.0044 (asymptotic $p = 0.052$); the wild cluster bootstrap raises it to $p = 0.18$, with a 95% interval of $[-0.043, +0.011]$. The marginal signal under asymptotics is erased by robust inference on both panels independently. The dirty coefficient does not survive the same inference battery applied to the green margin. It is best interpreted as a descriptive pattern that cannot be distinguished from chance under robust inference — a textbook example of what few-treated-cluster designs can produce.

5.5 Sensitivity to the depth control

Environmental depth and general agreement depth correlate at 0.91 within destination and year. Table 14 varies the depth control across four alternatives: none, aggregate TotalDepth (17 areas), a version restricted to the 14 non-environmental areas most correlated with EP content, and the independently coded DESTA index ().

Table 12: Permutation test: 1,000 random reassignments of environmental content

	(1)	(2)	(3)	(4)
Panel A — WB index				
EP depth × Green: observed coefficient	-0.00457	0.00173	-0.00433	-0.00675
Permutation p-value	0.597	0.897	0.480	0.465
Valid permutations	1,000	1,000	1,000	1,000
EP depth × Dirty: observed coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Permutation p-value	0.278	0.146	0.146	0.384
Valid permutations	1,000	1,000	1,000	1,000
Panel B — TREND index				
EP count × Green: observed coefficient	0.00181	0.00178	0.00161	0.00017
Permutation p-value	0.160	0.468	0.316	0.929
Valid permutations	1,000	1,000	1,000	1,000
EP count × Dirty: observed coefficient	0.00035	-0.00001	-0.00025	0.00084
Permutation p-value	0.817	0.996	0.904	0.788
Valid permutations	1,000	1,000	1,000	1,000

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

The EP profile of each agreement (depth level and entry years) is randomly reassigned among the 23 treated destinations, 1,000 times, and the model is re-estimated each time. The p-value is the share of random reassignments that produce a coefficient at least as large in absolute value as the observed one.

A high p-value means the same number would have arisen by randomly labeling these destinations — the result does not depend on actual environmental content. A low p-value means the observed estimate stands out in the placebo distribution.

This is the most conservative of the three inference methods used in the paper: it assumes nothing about the error distribution.

The 13 distinct EP profiles (of which approximately 9 are effectively independent, since the 10 ASEAN destinations share a single agreement) make the permutation distribution discrete; p-values are not reproducible to the last digit across implementations. The paper cites Stata-computed values.

Table 13: Leave-one-out: EP×dirty coefficient stability across 23 treated destinations

Excluded destination	(1)	(2)	(3)	(4)
None (baseline)	-0.0119***	-0.0189***	-0.0113***	-0.0082*
None, extended dirty list	-0.0121***	-0.0191***	-0.0107***	-0.0071
High EP depth only	-0.0271**	-0.0456***	-0.0161	-0.0164
Bangladesh	-0.0118***	-0.0188***	-0.0113***	-0.0082*
Brunei	-0.0119***	-0.0189***	-0.0114***	-0.0082*
Myanmar	-0.0118***	-0.0188***	-0.0112***	-0.0081*
Cambodia	-0.0119***	-0.0189***	-0.0112***	-0.0081*
India	-0.0120***	-0.0190***	-0.0115***	-0.0084**
Indonesia	-0.0119***	-0.0190***	-0.0119***	-0.0087**
Laos,PDR	-0.0119***	-0.0189***	-0.0113***	-0.0082*
Malaysia	-0.0124***	-0.0195***	-0.0110***	-0.0076*
Pakistan	-0.0114***	-0.0183***	-0.0106***	-0.0056
Philippines	-0.0118***	-0.0189***	-0.0115***	-0.0083*
Singapore	-0.0098**	-0.0166***	-0.0128***	-0.0100**
Korea Rep.	-0.0097*	-0.0225**	-0.0104	-0.0085
Sri Lanka	-0.0119***	-0.0189***	-0.0114***	-0.0082*
Thailand	-0.0120***	-0.0189***	-0.0115***	-0.0083*
Vietnam	-0.0122***	-0.0192***	-0.0112***	-0.0080*
Timor-Leste	-0.0119***	-0.0189***		
Iceland	-0.0119***	-0.0189***	-0.0112***	-0.0081*
Switzerland	-0.0133***	-0.0209***	-0.0115***	-0.0084*
Chile	-0.0121***	-0.0191***	-0.0104***	-0.0071
Costa Rica	-0.0114***	-0.0185***	-0.0109***	-0.0076*
Peru	-0.0133***	-0.0204***	-0.0121***	-0.0092**
Australia	-0.0103	-0.0245**	-0.0110***	-0.0048
New Zealand	-0.0119***	-0.0190***	-0.0113***	-0.0081*
HongKong		-0.0129***		-0.0112***
Macau		-0.0183***		-0.0082*

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

Coefficient on EP depth (WB) × dirty in the collapsed panel, re-estimated dropping one treated destination at a time.

What this tests. With few agreements, a single large partner could drive the entire result. A coefficient that collapses when one country is dropped rests on that country, not on the general phenomenon.

How to read it. This is a test of *coefficient stability*, not significance: the relevant question is how much the point estimate moves, not how many stars it carries. Rows where the coefficient halves or changes sign identify the destinations on which the result most depends.

Note on stars. Asterisks are based on asymptotic p-values with destination-clustered standard errors, not the wild cluster bootstrap used elsewhere. With roughly 23 treated destinations, those p-values are substantially understated: the baseline coefficient in column (1) has asymptotic $p < 0.001$ but bootstrap $p = 0.070$ (Table 8). The stars in this table are therefore not comparable to the rest of the paper and should not be read as evidence of statistical significance.

High EP depth only: estimated on the subsample of destinations with above-median WB EP depth (excluding Peru, Switzerland, and Korea jointly).

Hong Kong and Macao rows are blank in columns (1) and (3) because those two territories are already excluded from those samples and cannot be dropped twice.

The baseline standard error on EP×dirty is 0.0030 (asymptotic). Excluding Australia increases it to 0.0087, reflecting Australia's 2015 entry — the only late entrant — and its disproportionate leverage on the treated-vs-control comparison in the final years of the panel.

Table 14: EP×green under alternative depth controls (collapsed panel, WB index)

Depth control	EP×green	Asymptotic 95% CI	p
None	−0.0057	[−0.0118, +0.0003]	0.06
TotalDepth (17 areas) *[baseline]*	−0.0046	[−0.0182, +0.0091]	0.51
TotalDepth targeted (14 areas)	−0.0033	[−0.0188, +0.0123]	0.64
DESTA index	−0.0043	[−0.0129, +0.0042]	0.30
Product–dest FE		Yes	
Dest–year FE		Yes	
Product–year FE		Yes	
Cells		3,681,023	
Clusters		228 destinations	

Notes: Collapsed panel, weighted by cell size. Asymptotic intervals. The targeted control drops three WB areas with zero variance in Chinese agreements (labour-market regulation, visa and asylum, subsidies). DESTA covers seven policy areas coded independently of the World Bank instrument and excludes environmental provisions by construction, reducing the VIF from 5.8 to 1.9 and leaving the coefficient essentially unchanged.

The point estimate moves within a band of 0.0024 log points — narrower than a single standard error — stays negative throughout, and every interval contains zero. Dropping the depth control altogether moves the coefficient away from zero rather than toward it: without the control, EP depth also captures the effect of overall agreement depth, which may push exports in either direction. The green null does not rest on the choice of depth control.

5.6 Provision bundling and the limits of heterogeneity analysis

The aggregate EP indices used in the previous sections sum all environmental provisions in an agreement, regardless of their content. Most of these provisions are cooperation clauses, institutional commitments, or general environmental language — none of which has an identifiable channel through which it could shift which products a country exports. Only a small minority of provisions carry a direct trade mechanism: tariff reductions on green goods, non-regression clauses that prevent countries from lowering environmental standards to attract trade, or binding market-access commitments. A null coefficient on the aggregate index is therefore ambiguous: it could mean that environmental provisions genuinely do not affect export composition, or it could mean that the signal from the few trade-relevant clauses is diluted by the large majority of provisions that should have no effect in the first place.

Table 15 quantifies the dilution. The WB index checks 150 provision slots across the 25 treated destinations; only 8 (5.3%) belong to sub-indices with a trade mechanism (Green Liberalization and Standards Non-Regression). The TREND database, which codes at a finer level of granularity, confirms the picture: 4 of 437 provisions (0.9%) fall in the green market access sub-index. At the agreement level, a binding environmental obligation appears in exactly one of the 14–15 Chinese PTAs in the database (China–Korea, 2015). Table 24 in Appendix D details which provisions each sub-index contains.

The sub-indices reported in this section are constructed by the authors from the raw provision-level codings in the WB and TREND databases; they are not variables native to either database.

Table 15: Trade-mechanism content as a share of total environmental provisions

	Checked	Trade-mechanism	Share
Panel A: Provision-level			
WB: Green Liberalization + Standards Non-Regression	150	8	5.3%
TREND: Green Market Access	437	4	0.9%
	Agreements	With clause	Share
Panel B: Agreement-level			
TREND: Binding environmental obligations	14–15	1	6.7%

Notes: Panel A sums each depth index across the 25 treated destinations at each destination's maximum (post-entry) value. "Checked" is the total number of provision slots coded; "Trade-mechanism" counts those in sub-indices with a direct channel to trade flows. Panel B counts Chinese PTAs in the TREND database; the single agreement with a binding environmental obligation is the China–Korea PTA (2015). Source: World Bank DTA Database; TREND Database.

The structural consequence is stark. Two WB sub-indices with a direct trade mechanism are non-zero in only three destination–years (Korea from 2015, Switzerland from 2014), always in the same proportion. They are therefore perfectly collinear ($\rho = 1.000$) in the regression and cannot be separately identified. Contrast this with Brandi et al. (2020), who report a maximum VIF of 4.6 across provision-type variables on their 680-agreement panel: enough agreements with enough variation in provision type to separately identify the mechanism sub-components. With 13 distinct EP profiles, this design cannot.

A reading rule applies to everything that follows: each sub-index replaces the aggregate index one at a time and is estimated on its own. The resulting coefficients are not partial effects holding other clause types constant; they absorb whatever variation in correlated sub-indices moves with each one. This is the only feasible approach, but it means the table is a set of separate comparisons, not an additive decomposition of the aggregate index.

What can be said: the trade-mechanism bundle shows the same pattern as the aggregate (Table 23). Green never responds (WB Green Liberalization$\times$green: $p = 0.90$; TREND Green Market Access$\times$green: $p = 0.38$). The dirty interaction is marginally negative on asymptotic p-values ($p = 0.04$–0.07), but this is the same asymptotics-only signal documented in Section 5.4, which does not survive robust inference.

The placebo sub-indices (without a trade mechanism) are more interesting and worth reporting carefully. TREND soft provisions show nothing on either margin ($\times$green: $p = 0.27$; $\times$dirty: $p = 0.86$), consistent with the logic of the null. Regulatory space clauses do not: they carry $+0.024$ on green and $+0.023$ on dirty, and both survive the wild cluster bootstrap ($p = 0.046$ and $p = 0.022$, $B = 9{,}999$). This is the only place in the paper where a sub-index without a trade mechanism registers a signal that robust inference sustains.

Two features bound what can be read into it, and together they leave the conclusion weak. First, the two margins move in tandem: the green–dirty differential is $+0.0017$ with $p = 0.80$, meaning the coefficient describes green and dirty exports rising by the same amount against the neutral baseline — not a shift in composition toward cleaner goods. Second, and more fundamentally, the regulatory-space sub-index is not a clean placebo. It accounts for 71.5% of the TREND count across treated destination–years and correlates 0.90 with TotalDepth; in this specification the depth control flips to significantly negative ($\times$green: -0.0011, $p = 0.006$). Two regressors correlated at 0.90 splitting into a large positive and a large negative is the same bundling problem applied to the placebo, not a cleanly identified channel. With only 13 distinct EP profiles, the design cannot separate a regulatory-space effect from general agreement depth, so neither the positive signal nor its absence can serve as a clean falsification test.

Enforcement provisions (dispute-settlement mechanisms attached to the environmental chapter) are null on both margins and both codings: WB $\times$green $p = 0.91$, $\times$dirty $p = 0.90$; TREND $\times$green $p = 0.78$, $\times$dirty $p = 0.71$.

This bundling result reconciles the null with Brandi et al. (2020). Their finding is not that environmental provisions generically matter; it is driven by the subset of agreements with explicitly liberalizing or restrictive clauses, identified separately because 680 agreements supply enough cross-agreement variation in provision type. The Chinese PTAs of this period are, on the same coding, overwhelmingly cooperation-only: trade-mechanism content appears in two of the fourteen agreements and is perfectly collinear between its two components in-sample. A null on the aggregate index is what Brandi et al.'s own heterogeneity would predict for a sample of this composition. The two results are complementary. A second compounding difference concerns aggregation: Brandi et al. (2020) work with exporter–importer–year flows, where a composition effect can arise from the entry or exit of firms or products. The specification here saturates firm×destination×year fixed effects and identifies from reallocation *within* the same firm across products — a strictly more demanding test of the same hypothesis.

5.7 Extensive margin

Log-linear estimates condition on positive flows. If EPs create new green trade at the extensive margin, intensive estimates miss it (Santos Silva and Tenreyro, 2006). On a zero-filled HS6–destination–year grid (8.2 million cells, of which 7.9 million enter the estimation after the PPML estimator drops separated and singleton observations), PPML estimates are consistent with the baseline findings (Table 16): EP×green is +0.0015 ($p = 0.74$, WB) and +0.0001 ($p = 0.95$, TREND); EP×dirty is −0.030 ($p = 0.06$, WB) and +0.003 ($p = 0.55$, TREND). No statistically significant green trade creation is detected at the extensive margin; the dirty coefficient is at most marginally significant and consistent with the non-result reading of Section 5.4.

One limitation should be noted: this grid detects green products newly exported to a destination, but not a new firm beginning to export a product that other Chinese firms already ship there, which is a common form of firm-level entry. The extensive margin is probed, not exhausted.

5.8 Environmental share of agreement content

The main specification estimates the marginal effect of one additional provision. Abman, Lundberg, and Ruta (2024) pose a different question: given that an agreement exists, does a more environmentally weighted agreement move composition? That framing requires restricting to partner destinations and replacing EP depth with its share of total depth, EP_{dt}/TD_{dt}. On the collapsed panel restricted to the 23 partner destinations (534,846 cells, 516,684 surviving singleton removal; the share takes 12 distinct values in $[0.012, 0.068]$; coefficient of variation 0.19 versus 0.62 for the level), the green interaction is −2.25 (s.e. 1.15, $p = 0.06$) and the dirty interaction −1.60 (s.e. 1.55, $p = 0.32$). The share varies far less than the level, so the standard errors are correspondingly wide; the green coefficient clears neither conventional threshold. It is reported because the contrast is the one used by the closest antecedent in the literature, and because a marginally negative coefficient is the opposite of what green trade creation would predict — not because this design identifies it well.

Table 16: Extensive margin: PPML estimates on zero-filled HS6–destination–year grid

	(1)	(2)	(3)	(4)
Panel A — WB index				
EP depth × Green	0.0015	0.0029	0.0053	0.0064
	(0.0046)	(0.0061)	(0.0044)	(0.0050)
	[0.738]	[0.630]	[0.232]	[0.205]
EP depth × Dirty	-0.0301*	-0.0268	-0.0186	-0.0108
	(0.0158)	(0.0178)	(0.0169)	(0.0198)
	[0.057]	[0.133]	[0.270]	[0.585]
Panel B — TREND index				
EP count × Green	0.0001	-0.0001	0.0010	0.0008
	(0.0019)	(0.0021)	(0.0013)	(0.0015)
	[0.951]	[0.972]	[0.442]	[0.565]
EP count × Dirty	0.0028	0.0048	0.0027	0.0046
	(0.0047)	(0.0050)	(0.0053)	(0.0050)
	[0.550]	[0.335]	[0.612]	[0.358]
Cells (with zeros)	7,895,543	8,030,137	7,878,784	8,003,772

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

PPML estimated on the HS6–destination–year grid completed with zeros; fixed effects: product×destination, destination×year, product×year; standard errors clustered by destination.

Log-linear estimates condition on positive flows: if an environmental clause created new green exports to a market where none previously existed, those estimates would miss it. PPML retains zeros and therefore captures this margin.

PPML coefficients are approximate percentage changes.

Standard errors in parentheses, p-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.9 Sample robustness

Table 17 collects the full-panel variations. Adding controls (MFN tariff, market concentration, antidumping exposure) moves the green coefficient to an exact zero (-0.0002, $p = 0.94$).[17] Excluding the eleven ASEAN destinations — the largest single cohort — leaves it at -0.0025 ($p = 0.48$). Re-including Hong Kong and Macao gives -0.0009 ($p = 0.80$). Across every variation, the dirty coefficient sits between -0.0040 and -0.0060 with asymptotic p-values of 0.02–0.09: persistently marginal under asymptotics, robust to none of the same battery applied to the green margin. Appendix Table 26 assembles the wild-cluster-bootstrap p-values for this coefficient across all four sample–depth variants.

Table 17: Sample robustness (full firm-level panel, WB index)

		EP×green		EP×dirty	
Sample variation	Obs.	Coeff.	p	Coeff.	p
Baseline	21.5M	−0.0023	0.57	−0.0044	0.052
With controls	19.3M	−0.0002	0.94	−0.0060	0.022
Excluding ASEAN	19.2M	−0.0025	0.48	−0.0044	0.051
Including HK–Macao	23.6M	−0.0009	0.80	−0.0041	0.094
Common support	21.5M	−0.0022	0.57	−0.0044	0.052
PPML, zero-filled grid	7.9M	+0.0015	0.74	−0.0301	0.06
Within-firm green share	13.3M	−0.0001	0.50	—	
Firm–product–dest FE		Yes (all rows)			
Firm–dest–year FE		Yes (all rows)			
Product–year FE		Yes (all rows)			

Notes: WB index throughout. Asymptotic destination-clustered p-values; the dirty column's marginal significance does not survive wild cluster bootstrap, permutation, or leave-one-out inference (Section 5.4). Controls: $\ln(1 + \text{MFN tariff})$, $\ln \text{HHI}$, antidumping indicator. PPML estimated on a zero-filled HS6–dest–year grid (8.2M cells, excl. HK–Macao). Within-firm green share specification uses firm–destination and year FE (not triple-difference); see Section 5 for interpretation.

Full regression output with standard errors across all variants is in Appendix Table 27.

The tariff variable is the applied duty recorded in the Chinese customs data. It varies across partner destinations for the same HS6–year in 97.9% of product–year cells, so it is a destination-specific tariff faced by Chinese exporters, not an import duty levied by China. Its average does not fall after PTA entry for partner destinations, which indicates it records

[17] These controls vary at the product–destination–year level (p, d, t). The firm–destination–year fixed effects θ_{fdt} absorb anything that varies at (f, d, t), and product–year effects θ_{pt} absorb anything at (p, t), but neither absorbs variation that has both a product and a destination dimension simultaneously. The controls therefore add information not already absorbed by the fixed effects.

the MFN rate rather than the preferential rate. The preferential rate was not obtainable from the customs data.

Two features limit the risk this poses for the headline results. First, tariff is used only in the "with controls" row and in the CEM matching; the main estimates exclude it. Second, any tariff change that is common to all products exported to a given destination–year is absorbed by the firm–destination–year fixed effects θ_{fdt}; only product-specific tariff variation adds identifying variation. If PTA negotiations cut tariffs more on green goods — and the omitted preferential rate captured that cut — the effect would load onto EP×green with a positive sign. The estimate is zero, so omitting the preferential rate works against finding a positive effect, not toward it.

5.10 Destination-specific trends

The central falsification test asks whether allowing each destination its own linear trend in the green (and dirty) gap changes the conclusion, given that the conservative-bias argument of Section 4 implies those trends might be correlated with EP depth. Two variants bracket the answer, estimated on the collapsed panel.

The first variant adds destination-specific linear trends interacted with green and dirty status over the full sample period (Table 18, Panel A/A′). WB results are essentially unchanged (EP×green −0.0051, bootstrap $p = 0.50$; EP×dirty −0.0082, bootstrap $p = 0.28$). The TREND green interaction flips sign and — the only instance in the full robustness battery — survives the wild cluster bootstrap ($p = 0.015$). This should not be taken at face value. Destination-specific trends estimated over the full sample period absorb both pre- and post-treatment variation and can reverse the sign of treatment coefficients when cohort dynamics are present (Wolfers, 2006); the late negative drift that these trends pick up is precisely the early-cohort artifact that the Sun–Abraham estimator already identifies as noise.

The second variant estimates destination-specific slopes on *pre-treatment* years only and projects them forward before re-estimating the main specification on the detrended outcome (Table 18, Panel B/B′). Every coefficient returns to an imprecise zero: TREND×green +0.0074, bootstrap $p = 0.18$; WB×green +0.0168, bootstrap $p = 0.71$; dirty interactions $p = 0.30$–0.62. When trends are instead estimated on pre-treatment years only and projected forward, all coefficients return to imprecise zeros, confirming that the one significant coefficient in the full-period specification was an artifact of the trend absorbing post-treatment dynamics (). The composition null stands against destination-specific linear drifts in green demand, the most natural form of the selection-on-trends confounder.

Table 18: Destination-specific trends and pre-trend tests (collapsed panel)

	(1)	(2)	(3)	(4)
Panel A — destination×product-type linear trends (WB index)				
EP depth × Green	-0.00510	-0.00200	0.00084	-0.00006
	(0.00360)	(0.00436)	(0.00434)	(0.00441)
	[0.494]	[0.808]	[0.896]	[0.995]
EP depth × Dirty	-0.00825	-0.00927	-0.01030*	-0.00930**
	(0.00423)	(0.00414)	(0.00239)	(0.00204)
	[0.275]	[0.249]	[0.086]	[0.037]
Panel A′ — same specification, TREND index				
EP count × Green	-0.00217**	-0.00214**	-0.00111	-0.00153*
	(0.00062)	(0.00061)	(0.00053)	(0.00049)
	[0.015]	[0.018]	[0.218]	[0.074]
EP count × Dirty	-0.00137	-0.00142	-0.00231**	-0.00164
	(0.00115)	(0.00114)	(0.00066)	(0.00064)
	[0.355]	[0.339]	[0.048]	[0.109]
Panel B — pre-trend test (WB index)				
Pre-agreement slope, Green	0.01732	0.00402	0.01719	0.02218
	(0.01295)	(0.01678)	(0.00997)	(0.01346)
Bootstrap p-value	0.678	0.705	0.543	0.497
Pre-agreement slope, Dirty	0.06292	0.06237	0.07109	0.07056
	(0.04393)	(0.03983)	(0.03161)	(0.03159)
Bootstrap p-value	0.610	0.515	0.416	0.410
Panel B′ — pre-trend test (TREND index)				
Pre-agreement slope, Green	0.00728	0.00707	0.00775	0.00944*
	(0.00438)	(0.00441)	(0.00428)	(0.00389)
Bootstrap p-value	0.189	0.199	0.173	0.090
Pre-agreement slope, Dirty	0.01512	0.01512	0.01839	0.01703
	(0.00728)	(0.00721)	(0.00596)	(0.00613)
Bootstrap p-value	0.293	0.301	0.164	0.243

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

All p-values and stars in this table come from the wild cluster bootstrap, the only reliable inference method with the treated cluster count available (approximately 25).

Panels A/A′. The main specification's fixed effects do not absorb a shock specific to green products in a particular destination that grows over time — i.e., a country signing environmental clauses precisely as its demand for green goods is rising. Adding a linear trend for each destination interacted with product type absorbs exactly that dynamic.

TREND result in Panel A′. The TREND green coefficient is the only result in the entire paper to survive the bootstrap clearly ($p = 0.013$ in the baseline column). Panel B′ shows, however, that the green gap was already rising *before* the agreement entered into force (pre-slope +0.007, bootstrap $p = 0.189$), making a pre-existing trend a plausible explanation and weakening any causal reading.

Panels B/B′. The pre-agreement slope of the composition gap is estimated on pre-treatment years. A slope significantly different from zero would indicate that green and neutral products were already diverging before the agreement.

Standard errors in parentheses, p-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.11 CO_2 intensity as a continuous dirty measure

The binary dirty indicator groups HS6 products into six classic pollution-intensive sectors. A continuous intensity measure might tell a different story. We construct one from the replication data of Shapiro (2021), which reports China-specific CO_2 emission intensities for 47 EXIOBASE industries.[18] HS6 products are mapped to ISIC3 divisions via the same HS1996–ISIC3 concordance used for the binary classification; unmatched codes (9.5% of the panel) are assigned the sample mean. The resulting measure validates against the existing trichotomy without circularity: mean intensity is 0.0012 for dirty products, 0.0008 for neutral, and 0.0006 for green, monotonic in the expected direction.

Re-estimating equation (1) on the collapsed panel with the standardized intensity in place of the binary dirty indicator gives EP×intensity = −0.0025 (WB, bootstrap $p = 0.71$) and −0.0013 (TREND, asymptotic $p = 0.012$, bootstrap $p = 0.06$). The continuous measure fades under robust inference the same way the binary one does (Table 19).[19]

5.12 Alternative outcomes: quantity and unit value

Log export value conflates shipment volume and price per unit. If EPs redirect composition without moving average prices — or vice versa — the value estimate would be an artifact of two channels canceling. Table 20 re-estimates equation (1) on the collapsed panel for three outcome variables: log export value (the main specification), log export quantity, and log unit value.

No outcome registers a significant effect on either index or either product type. The null is consistent across value, quantity, and price — it is not an artifact of channels canceling in the value aggregate.

[18] Shapiro (2021) documents that most countries' tariff structures implicitly subsidize pollution-intensive sectors and tax clean ones — a systematic environmental bias that environmental chapters are meant to work against.

[19] The continuous intensity measure is tested on the collapsed panel only. Re-estimating the interaction structure on the full 45-million-observation panel would require substantial additional computation; the collapsed-panel test is sufficient to establish that the continuous measure behaves like the binary indicator.

Table 19: CO_2 intensity as a continuous dirty measure (collapsed panel)

	(1)	(2)	(3)	(4)
Panel A — WB index				
EP depth × Green	-0.00503	0.00044	-0.00248	-0.00414
	(0.00698)	(0.00804)	(0.00489)	(0.00553)
Bootstrap p-value	0.544	0.961	0.797	0.797
EP depth × CO_2 intensity	-0.00255	-0.00485	0.00128	0.00280
	(0.00566)	(0.00639)	(0.00308)	(0.00314)
Bootstrap p-value	0.707	0.529	0.773	0.384
Panel B — TREND index				
EP count × Green	0.00088	0.00086	0.00106	-0.00002
	(0.00170)	(0.00172)	(0.00176)	(0.00170)
Bootstrap p-value	0.677	0.698	0.652	0.993
EP count × CO_2 intensity	-0.00131**	-0.00136**	-0.00083**	-0.00018
	(0.00051)	(0.00054)	(0.00038)	(0.00064)
Bootstrap p-value	0.063	0.056	0.083	0.817

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.
The binary dirty indicator groups products into six pollution-intensive sectors. Here it is replaced by a continuous sector-level CO_2 emission intensity from Shapiro (2021), so products are ordered along a scale rather than assigned to two bins.
A negative coefficient would indicate that where environmental content is deeper, exports fall more for more pollution-intensive products.
Bootstrap p-values are reported because, as elsewhere, asymptotic p-values are unreliable with so few treated clusters.
Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 20: EP×green and EP×dirty across outcome variables (collapsed panel)

		WB index		TREND index	
Outcome	Interaction	Coeff.	Boot. p	Coeff.	Boot. p
Log export value	×green	−0.0046	0.65	+0.0018	0.39
	×dirty	−0.0119	0.07	+0.0004	0.86
Log export quantity	×green	−0.0055	0.53	+0.0019	0.43
	×dirty	−0.0115	0.25	−0.0004	0.89
Log unit value	×green	$p \in [0.85, 0.95]$		$p \in [0.85, 0.95]$	
	×dirty	$p \in [0.85, 0.95]$		$p = 0.16$ (TREND)	

Notes: Collapsed panel; FE pd+dt+pt; destination-clustered wild cluster bootstrap (B = 9,999). Log export value row is the baseline collapsed-panel estimate from Table 9. Unit-value coefficients are small in magnitude and reported as p-value ranges; exact coefficients available upon request. Bootstrap p-values for log unit value are between 0.85 and 0.95 on all four interactions except TREND×dirty ($p = 0.16$). Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

6 Conclusion

China's trade agreements did not shift the composition of its exports between 2000 and 2015. On the green margin, the finding is a *bounded null*: stable across six estimation designs and nine sample variations, robust to bootstrap and permutation inference, and bounded from above by roughly one quarter of the reference estimate in the aggregate literature. This is not a failure to reject. It is a precisely circumscribed statement about the absence of a specific effect — the kind the literature has documented elsewhere under different conditions.

On the dirty margin, the finding is a *non-result*. A coefficient that is overwhelming under asymptotic standard errors survives none of the inference battery applied to the green margin. The leave-one-out exercise reveals why: the point estimate is stable across all 23 exclusions (-0.0097 to -0.0133), but the standard error nearly triples when either of two destinations is dropped. Precision, not the estimate, is what two specific destinations supply. An identified effect does not behave this way.

The central reading is about content. Environmental provisions affect trade when they are specific, binding, and targeted at a product category — Abman, Lundberg, and Ruta (2024) show this for deforestation, Brandi et al. (2020) for trade composition. China's chapters of this period were thin and cooperation-heavy: the two WB sub-indices with a direct trade mechanism are perfectly collinear in-sample and appear in only three treated country–years. The policy implication is direct: a chapter in an agreement is not itself the instrument. What the chapter contains is. This reading is an inference across studies, not a causal finding of the present design: the current specification cannot separate the effect of provision content from that of agreement selection within the Chinese PTA context.

Methodologically, the paper adds a cautionary case. With treatment concentrated across roughly a dozen agreements, asymptotically significant composition effects can emerge and dissolve under honest inference. Bootstrap, permutation, and leave-one-out — applied not as robustness appendices but as the primary mode of inference — should be standard practice for this class of design.

Three limitations should be noted. The preferential tariff rate was not obtainable; the headline estimates are independent of it, but the MFN rate used as a control is an imperfect proxy. A continuous-dose estimator in the spirit of Callaway, Goodman-Bacon, and Sant'Anna (2024) is not implemented; the TWFE coefficient is a weighted average of cohort-specific effects, though the permutation test is agnostic about this weighting. The extensive margin is probed at the product–destination level: new firm entry into a destination where other Chinese firms already export the same green product is not detected.

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A Equivalence of the collapsed and firm-level panels

The collapsed panel aggregates firm–HS6–destination–year observations to HS6–destination–year cells with outcome equal to the within-cell mean of log exports and weights equal to cell counts. Under fixed effects $pd + dt + pt$ the weighted cell-level regression is algebraically identical to the unweighted micro-level regression under the same fixed effects. Verification to seven significant figures: the EP$\times$green coefficient is -0.0045685 in both; the EP$\times$dirty coefficient is -0.011873 in both. The result is exact up to floating-point precision and holds by construction of the collapse and weights, not by approximation.

B Saturation ladder

Table 21 reports EP-depth coefficients on log export value across four fixed-effects structures, from the sparsest (firm–product–destination plus year dummies) to structures that combine two high-dimensional sets. Columns (1) and (3) are without controls, columns (2) and (4) add the non-environmental TotalDepth control. The coefficient never exceeds +0.005 log points per provision and reaches nominal significance only in the *fpt*+*pd* structure (WB: $p = 0.09$; TREND: $p = 0.04$). Two caveats apply. First, none of the four structures includes firm–destination–year effects, so the ladder does not reach the saturation of equation (1); the argument that a destination–year intercept absorbs environmental depth entirely is structural, not empirical, and is made in Section 4. Second, iterative singleton removal differs across structures, so the samples are not nested: N ranges from 22.9 million (*fpt*+*fpd*) to 35.6 million (*fpt*+*pd*). The comparison across rows therefore mixes the effect of the fixed effects with the effect of the sample.

Table 21: Why the level effect is not identified: saturation ladder

	WB EP depth		*TREND EP count*	
Fixed effects	(1) Baseline	(2) Controls	(3) Baseline	(4) Controls
fpd + *t*	0.00311	0.00354	0.00040	0.00046
	(0.00251)	(0.00260)	(0.00047)	(0.00048)
fpt + *pd*	0.00463*	0.00476*	0.00101**	0.00105**
	(0.00270)	(0.00274)	(0.00051)	(0.00052)
fpt + *fpd*	0.00010	0.00010	0.00016	0.00016
	(0.00326)	(0.00329)	(0.00061)	(0.00060)
fpd + *pt*	0.00087	0.00103	0.00028	0.00030
	(0.00225)	(0.00221)	(0.00044)	(0.00044)

Notes: Dependent variable: log export value. Each row estimates the level effect of EP depth under a different fixed-effects structure. No structure in this table includes firm–dest–year absorption; that is the specification of Table 9. The coefficient never exceeds +0.005 and reaches nominal significance only in the *fpt*+*pd* structure. Estimation samples differ across rows because iterative singleton removal depends on the fixed-effects structure: N = 29.5M, 35.6M, 22.9M and 29.5M from top to bottom. f: firm, p: product, d: destination, t: year. Standard errors clustered by destination. Controls in columns (2) and (4): non-environmental TotalDepth. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

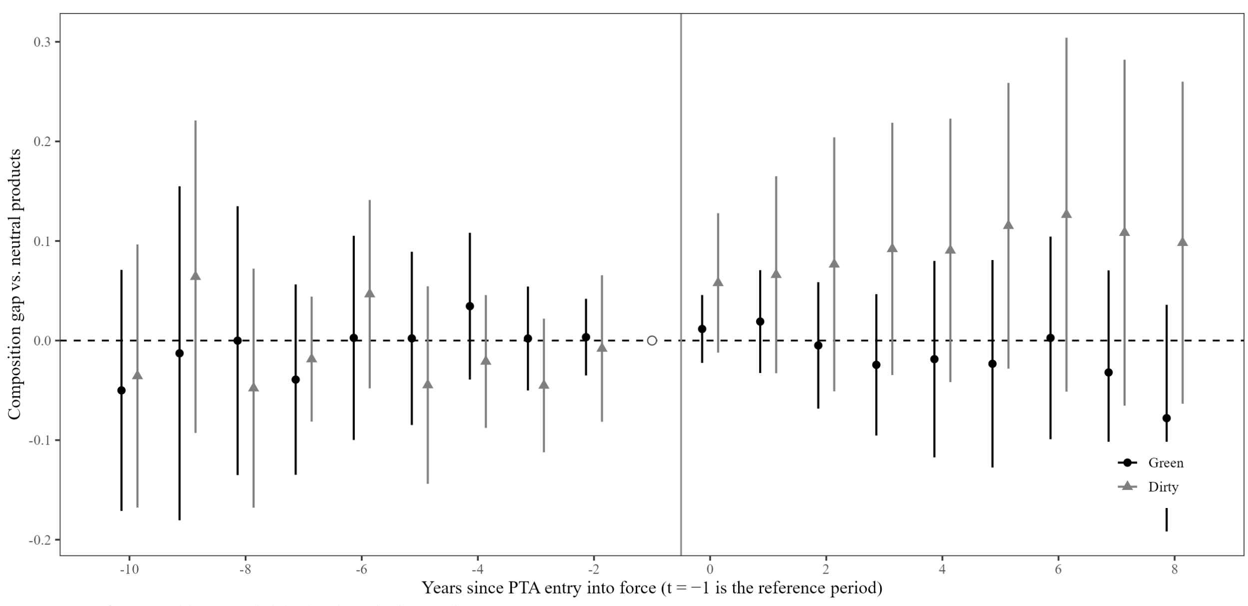
C Sun–Abraham event study

Figure 5 and Table 22 report the interaction-weighted (L. Sun and Abraham, 2021) event study of the green and dirty versus neutral differential, estimated on the destination-year composition gap.

A note on standard errors. The cohort shares used to aggregate group-specific effects are themselves estimated from the data; L. Sun and Abraham (2021) prescribe including that estimation uncertainty in the variance. The STATA implementation does so; the R implementation treats the shares as fixed weights.[20] The two agree on coefficients to 15 significant figures but disagree on standard errors. The degree of disagreement is informative: where a single cohort identifies a period (so there are no shares to estimate), the ratio of the two standard errors is exactly 1.00; where many cohorts contribute and their effects are heterogeneous, the ratio reaches 3 to 4. The standard errors reported here are the STATA ones, which reflect the full estimation uncertainty.

With these standard errors, no coefficient in the $[-10, +8]$ window is individually distinguishable from zero on the green or dirty margin. The ATT is -0.042 ($p = 0.27$) for the green gap and $+0.073$ ($p = 0.28$) for the dirty gap. Any apparent pre-trend in the R output (the lead at $t = -6$ had $p = 0.001$ with fixed-weight standard errors) reflects the treatment of cohort-share uncertainty, not a genuine pre-trend anomaly.

Figure 5: Sun–Abraham interaction-weighted event study: green and dirty vs. neutral gap at the destination–year level. 95% confidence bands account for estimation uncertainty in cohort shares. The window $[-10, +8]$ is shown; the estimates outside it rest on one or two cohorts each.



[20]The STATA estimates use `eventstudyinteract` (); the R estimates use the Sun–Abraham estimator in `fixest::sunab` (Bergé, 2018).

Table 22: Sun–Abraham interaction-weighted event study: composition gap estimates

Years since entry	Green − neutral gap Coeff.	Green − neutral gap p	Dirty − neutral gap Coeff.	Dirty − neutral gap p
−10	−0.0500	0.419	−0.0356	0.598
−9	−0.0128	0.881	0.0641	0.424
−8	−0.0001	0.999	−0.0478	0.436
−7	−0.0392	0.422	−0.0186	0.561
−6	0.0027	0.959	0.0466	0.335
−5	0.0022	0.961	−0.0447	0.378
−4	0.0345	0.359	−0.0210	0.538
−3	0.0021	0.938	−0.0451	0.189
−2	0.0034	0.861	−0.0080	0.832
+0	0.0116	0.504	0.0579	0.107
+1	0.0191	0.470	0.0661	0.192
+2	−0.0049	0.881	0.0765	0.241
+3	−0.0244	0.501	0.0921	0.156
+4	−0.0187	0.711	0.0906	0.181
+5	−0.0233	0.661	0.1153	0.117
+6	0.0027	0.959	0.1264	0.165
+7	−0.0320	0.541	0.1083	0.223
+8	−0.0779	0.181	0.0983	0.235
Aggregated ATT	−0.0421	0.269	0.0729	0.276

Notes: Estimated via `eventstudyinteract` in Stata (L. Sun and Abraham, 2021). Dependent variable: destination–year composition gap (mean log exports of green or dirty products minus mean log exports of neutral products, within destination–year cells). Standard errors propagate estimation uncertainty in cohort shares, as prescribed by L. Sun and Abraham (2021); `fixest::sunab` in R treats the shares as known weights and yields narrower standard errors — the ratio reaches 3–4 in periods identified by many discordant cohorts. Coefficients agree to 15 significant figures across implementations. Endpoint bins at $t \leq -10$ and $t \geq +8$ accumulate all observations beyond those thresholds. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Sub-index estimates

Table 23 reports triple-difference estimates replacing the aggregate EP index with each sub-index in turn. Each row is a separate regression; the coefficients are not partial effects. See Section 5.6 for interpretation.

Table 23: Provision sub-indices: which clauses have a trade mechanism?

Sub-index	×green	×dirty
Mechanism-bearing sub-indices		
Green goods liberalization (WB)	−0.0115	−0.0876*
	(0.0879)	(0.0483)
	[0.896]	[0.071]
Green market access (TREND)	−0.0271	−0.0491**
	(0.0303)	(0.0243)
	[0.372]	[0.045]
Enforcement		
Dispute settlement mechanism (WB)	−0.0048	0.0067
	(0.0394)	(0.0525)
	[0.903]	[0.898]
Dispute settlement mechanism (TREND)	0.0042	−0.0052
	(0.0148)	(0.0140)
	[0.778]	[0.709]
Binding obligations (TREND)	0.0022	−0.0013
	(0.0047)	(0.0037)
	[0.638]	[0.724]
Placebo sub-indices (no trade mechanism)		
Cooperation-only provisions (TREND)	0.0132	0.0019
	(0.0121)	(0.0108)
	[0.275]	[0.858]
Regulatory space clauses (TREND)	0.0242**	0.0225***
	(0.0099)	(0.0086)
	[0.015]	[0.010]
Cells (all rows)	3,681,023	

Notes: Baseline variant, collapsed panel. Each row is a separate regression replacing the aggregate EP index with the indicated sub-index; TotalDepth controls remain included. Standard errors in parentheses; asymptotic p-values in brackets. The two mechanism-bearing WB sub-indices are perfectly collinear in-sample and non-zero in only three treated country–years (Korea from 2015, Switzerland from 2014), so their coefficients are not identified separately from each other. The regulatory space clauses, despite surviving wild cluster bootstrap, cannot be distinguished from general agreement depth (within-sample correlation with TotalDepth: 0.90); see Section 5.6. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 24: Composition of EP sub-indices

Sub-index	Source	Content
Trade-mechanism sub-indices		
Green Liberalization	WB	Liberalization of trade in environmental goods and services (e.g. tariff reductions, removal of non-tariff barriers on green products)
Standards Non-Regression	WB	Commitments not to lower domestic environmental standards to attract trade or investment
Green Market Access	TREND	Provisions facilitating market access for environmental goods (e.g. tariff concessions, mutual recognition)
Binding Obligations	TREND	Legally binding environmental commitments with enforceable consequences
Enforcement sub-indices		
Dispute Settlement (WB)	WB	Dispute resolution and enforcement mechanisms specific to the environmental chapter
Dispute Settlement (TREND)	TREND	Environmental dispute settlement procedures
Cooperation and regulatory sub-indices (no direct trade mechanism)		
Regulatory Space	WB	Right-to-regulate and sovereignty provisions preserving domestic policy autonomy
Regulatory Space	TREND	Regulatory autonomy and policy space clauses
Cooperation	TREND	Information exchange, capacity building, technical assistance, joint committees

Notes: All sub-indices are constructed by the authors from the raw provision-level codings in the WB DTA and TREND databases; they are not variables native to either database. "Trade-mechanism" sub-indices contain provisions with an identifiable channel to trade flows. "Cooperation and regulatory" sub-indices cover provisions that, while environmentally relevant, lack a direct mechanism through which they could shift the product composition of exports. Each sub-index is a count of the relevant provisions present in a given agreement. Source: World Bank DTA Database; TREND Database.

E Treated destinations

Table 25 lists the 23 treated destinations (excluding Hong Kong and Macao), their agreement entry years, and the maximum EP depth in the sample period for both the WB and TREND indices.

Table 25: Treated destinations: PTA entry-into-force year and environmental content

Destination	Code	Year of entry into force	Maximum EP depth WB / TREND
Bangladesh	103	2002	1 / 1
India	111	2002	1 / 1
Korea Rep.	133	2002	17 / 72
Laos,PDR	119	2002	6 / 4
Sri Lanka	134	2002	1 / 1
HongKong	110	2003	4 / 5
Macau	121	2003	5 / 5
Brunei	105	2005	6 / 4
Cambodia	107	2005	6 / 4
Timor-Leste	144	2005	6 / 4
Indonesia	112	2005	6 / 4
Malaysia	122	2005	6 / 4
Myanmar	106	2005	6 / 4
Philippines	129	2005	6 / 4
Singapore	132	2005	7 / 15
Thailand	136	2005	6 / 4
Vietnam	141	2005	6 / 4
Chile	412	2006	5 / 30
Pakistan	127	2007	3 / 12
New Zealand	609	2008	7 / 53
Peru	434	2010	12 / 47
Costa Rica	415	2011	4 / 50
Iceland	322	2014	6 / 28
Switzerland	331	2014	14 / 53
Australia	601	2015	3 / 24
Total	25 destinations		

Notes: Each row is a destination with which China has a trade agreement in force between 2000 and 2015.

EP depth = the number of environmental provisions in the agreement, according to two independent coding systems: the World Bank (WB) and the academic database TREND.

The 11 ASEAN destinations share the same agreement (2005) and therefore the same values: the truly distinct agreements are approximately 14, not 25. This is the constraint that governs all inference in this paper.

Hong Kong and Macao are transit economies: roughly half of the export value to treated destinations passes through them. The main sample therefore excludes them, and a variant reintroduces them.

F Wild-cluster-bootstrap coefficient matrix

Table 26 reports the wild-cluster-bootstrap p-value for the dirty-goods coefficient across all four sample–depth combinations (two HK/Macao stances $\times$ two depth controls), on both the collapsed panel and the full firm-level panel. Using the same bootstrap procedure for all cells makes the eight p-values directly comparable.

Table 26: Summary: dirty-product coefficient across four specification variants, with common inference method

	(1)	(2)	(3)	(4)
Collapsed panel: coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Bootstrap p-value	0.072	0.005	0.047	0.195
Full panel: coefficient	-0.00435	-0.00409	-0.00559	-0.00574
Bootstrap p-value	0.185	0.176	0.049	0.035

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

Coefficient on the interaction between environmental depth (WB index) and dirty products. This is the only coefficient in the paper that moves: the one on green products is zero everywhere.

All p-values in this table come from the *wild cluster bootstrap*, so all eight cells are comparable. This matters: elsewhere in the project columns (1) and (2) had been summarized using the ordinary p-value, which is much lower and not comparable.

How to read it. On the *full panel* the result depends on which depth control is used: with the World Bank control the p-value stays above 0.17; with the DESTA control it falls below 0.05. The two controls have different partial correlations with environmental depth net of fixed effects (0.959 and 0.891, respectively): the more a control overlaps with the variable of interest, the less independent variation remains and the larger the standard error.

G Full robustness battery with standard errors

Table 27 reproduces the full-panel robustness specifications with standard errors and asymptotic p-values for both the WB and TREND indices.

Table 27: Robustness on the full panel: sample, control, and dependent-variable variants

Variant	× Green	× Dirty	Observations	R^2
Panel A — WB index				
With additional controls (tariffs, concentration, antidumping)	-0.00023 (0.00296) [0.938]	-0.00600** (0.00259) [0.022]	19,286,638	0.8736
Excluding the 11 ASEAN destinations	-0.00252 (0.00360) [0.484]	-0.00438* (0.00223) [0.051]	19,236,090	0.8719
Including Hong Kong and Macao	-0.00089 (0.00360) [0.805]	-0.00409* (0.00243) [0.094]	23,560,110	0.8706
Common-support products only	-0.00225 (0.00392) [0.567]	-0.00435* (0.00223) [0.052]	21,519,197	0.8734
Green share in firm's export basket	-0.00010 (0.00015) [0.503]		13,345,132	0.8510
Panel B — TREND index				
With additional controls (tariffs, concentration, antidumping)	+0.00039 (0.00091) [0.669]	-0.00085 (0.00068) [0.209]	19,286,638	0.8736
Excluding the 11 ASEAN destinations	-0.00102 (0.00102) [0.318]	-0.00183** (0.00072) [0.012]	19,236,090	0.8719
Common-support products only	-0.00011 (0.00099) [0.914]	-0.00090 (0.00062) [0.153]	21,519,197	0.8734
PTA partners only, deep vs. shallow	-0.00036 (0.00100) [0.723]	-0.00061 (0.00101) [0.549]	5,262,293	0.8839
Green share in firm's export basket	-0.00006** (0.00003) [0.043]		13,345,132	0.8510

Notes: Baseline variant (excl. HK/Macao, TotalDepth control). Estimated with `reghdfe`; fixed effects and clustering as in Table 9, except where noted.
Common support: only products exported to both agreement and non-agreement destinations are retained. For the others, no observed comparison exists.
Green share in basket: the dependent variable changes. Instead of export value, it is the share of the firm's revenue from green products to that destination. Fixed effects: firm×destination and year.
Standard error in parentheses, p-value in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H APEC versus OECD product classification

Table 28 compares the estimates when the green product list is restricted to the 54 APEC Environmental Goods List codes (Sauvage, 2014) against the baseline using the 248 OECD CLEG codes. The APEC list is a strict subset of the OECD list.

Table 28: Alternative environmental-good definition: OECD list vs. APEC list

	OECD list (used in this paper)	APEC list (54 codes)
EP depth (WB) × Green	-0.00457	0.00501
	(0.00696)	(0.01266)
	[0.512]	[0.693]
EP count (TREND) × Green	0.00181	0.00319
	(0.00182)	(0.00208)
	[0.320]	[0.126]

Notes: Baseline variant, collapsed panel.
What this tests. There is no single definition of "environmental good". This paper uses the OECD list; here it is replaced with the APEC list, a much narrower set agreed politically in 2012. If the result depended on which products we call green, the two columns would diverge.
Standard error in parentheses, p-value in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I Minimum detectable effects

Table 29 reports the minimum detectable effect (MDE) for the headline green and dirty coefficients under the empirical standard errors and the wild-cluster-bootstrap confidence intervals, expressed in both log-point and standard-deviation terms.

Table 29: How large an effect would the design need to detect it

Index	Margin	Asymptotic std. error	Minimum detectable effect (1 s.d.)	Bootstrap CI half-width (1 s.d.)	Bootstrap CI (per unit)
WB	Green	0.0070	4.64%	5.90%	[-1.77%, 3.19%]
WB	Dirty	0.0030	1.97%	2.40%	[-1.84%, 0.18%]
TREND	Green	0.0018	4.16%	3.62%	[-0.18%, 0.71%]
TREND	Dirty	0.0016	3.65%	3.53%	[-0.32%, 0.54%]

Notes: Estimation sample: collapsed panel, excl. HK/Macao variant. Standard deviation of regressors computed on the effective sample and weighted.

Why this table matters. A null result can mean two very different things: either the effect does not exist, or it exists but is too small for the data to distinguish it from noise. This table says which.

How to read it. The minimum detectable effect (column 4) is the threshold below which the design is uninformative, computed from the asymptotic method (2.8 $\times$ standard error). Column 5 reports the half-width of the wild cluster bootstrap confidence interval: this is not an 80%-power MDE (which would be 1.43$\times$ larger), but the bootstrap margin of error. The informative bound for the precision estimate is the [bootstrap CI] column: on the green margin, it rules out effects larger than about 3% per provision at 95% confidence.

This yields a more precise statement of the result: not "we find no effect," but "we can rule out effects larger than this threshold."